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(54) SYSTEMS AND METHODS FOR GAS MONITORING AND GAS LEAK PREVENTION

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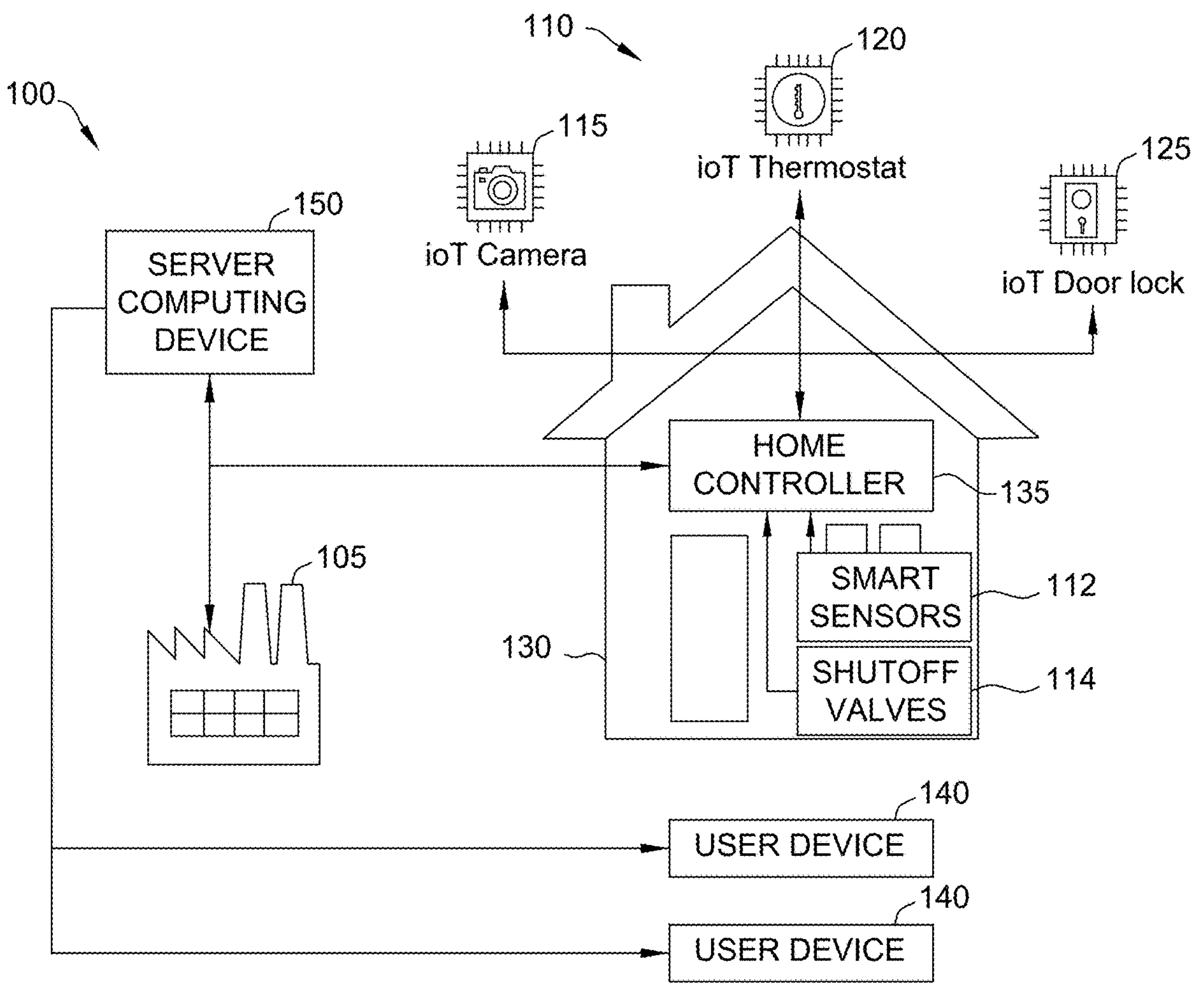
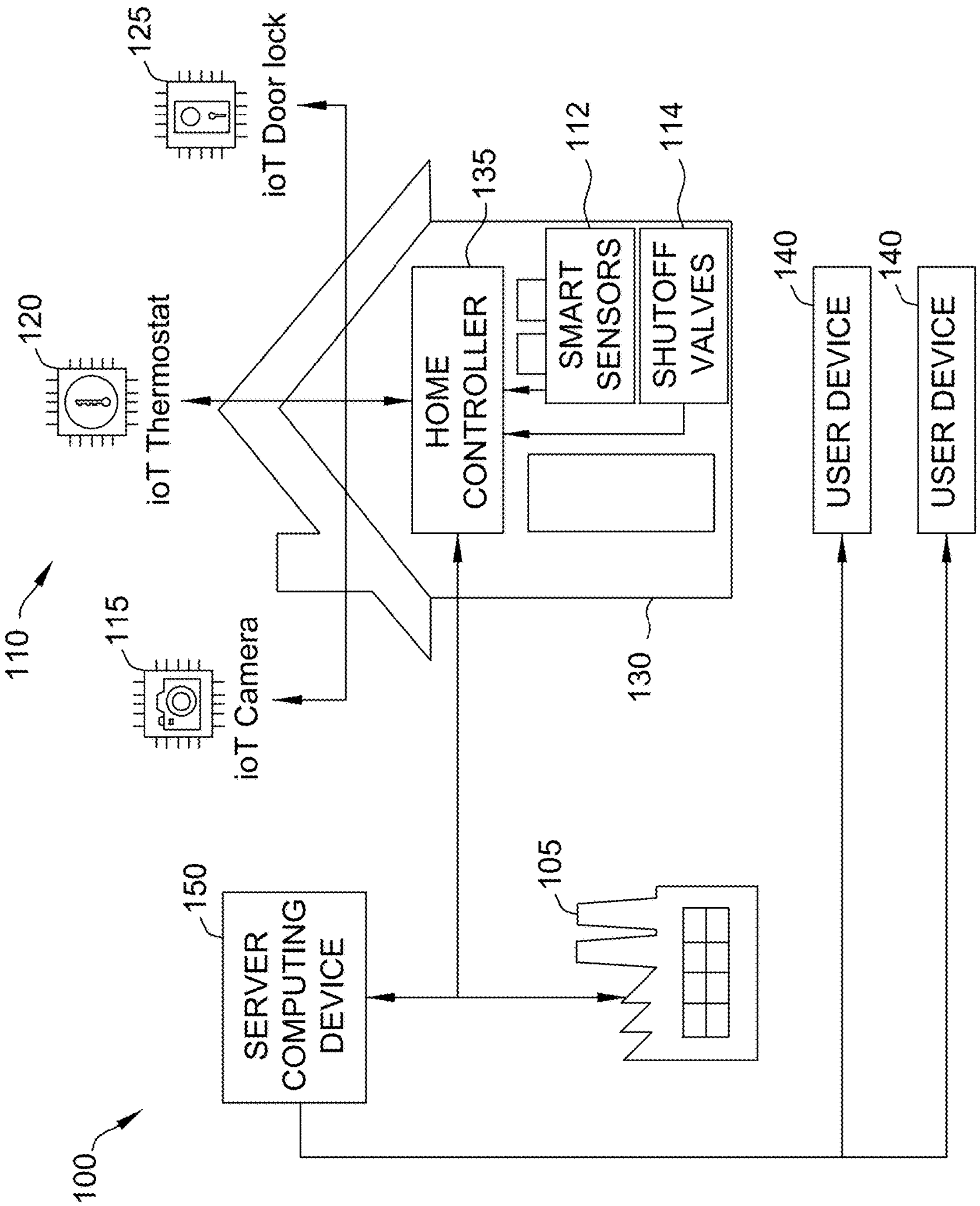
(57) **ABSTRACT**

A computer system is provided that may be programmed for home monitoring. The system may: (a) receive sensor data from at least one smart sensor; (b) compare the sensor data to one or more parameters generated using an artificial intelligence model to determine an abnormal level of gas is present in the home, wherein the artificial intelligence model is trained based upon historical sensor data relating to a plurality of homes; and/or (c) in response to determining that an abnormal level of gas is present in the home, control at least one gas shutoff valve to close to prevent further gas leaks at the home.





100
110
115
ioT Camera
120
ioT Thermostat
125
ioT Door lock
135
HOME CONTROLLER
112
SMART SENSORS
114
SHUTOFF VALVES
130
140
USER DEVICE
140
USER DEVICE
150
SERVER COMPUTING DEVICE
105

FIG. 1

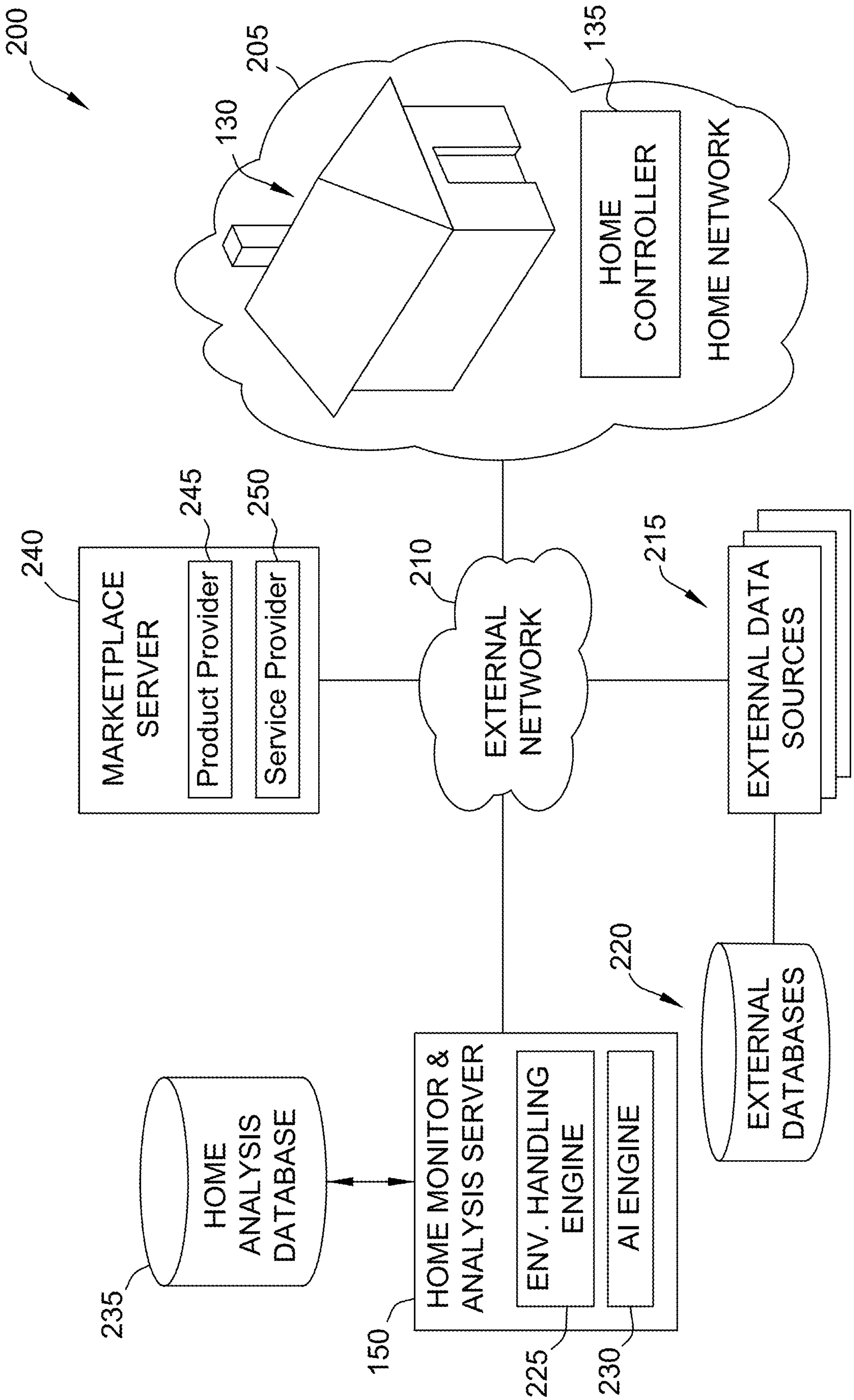

200
205
135
130
HOME CONTROLLER
HOME NETWORK
240
245
250
MARKETPLACE SERVER
Product Provider
Service Provider
210
EXTERNAL NETWORK
215
EXTERNAL DATA SOURCES
220
EXTERNAL DATABASES
235
HOME ANALYSIS DATABASE
150
HOME MONITOR & ANALYSIS SERVER
ENV. HANDLING ENGINE
AI ENGINE
225
230

FIG. 2

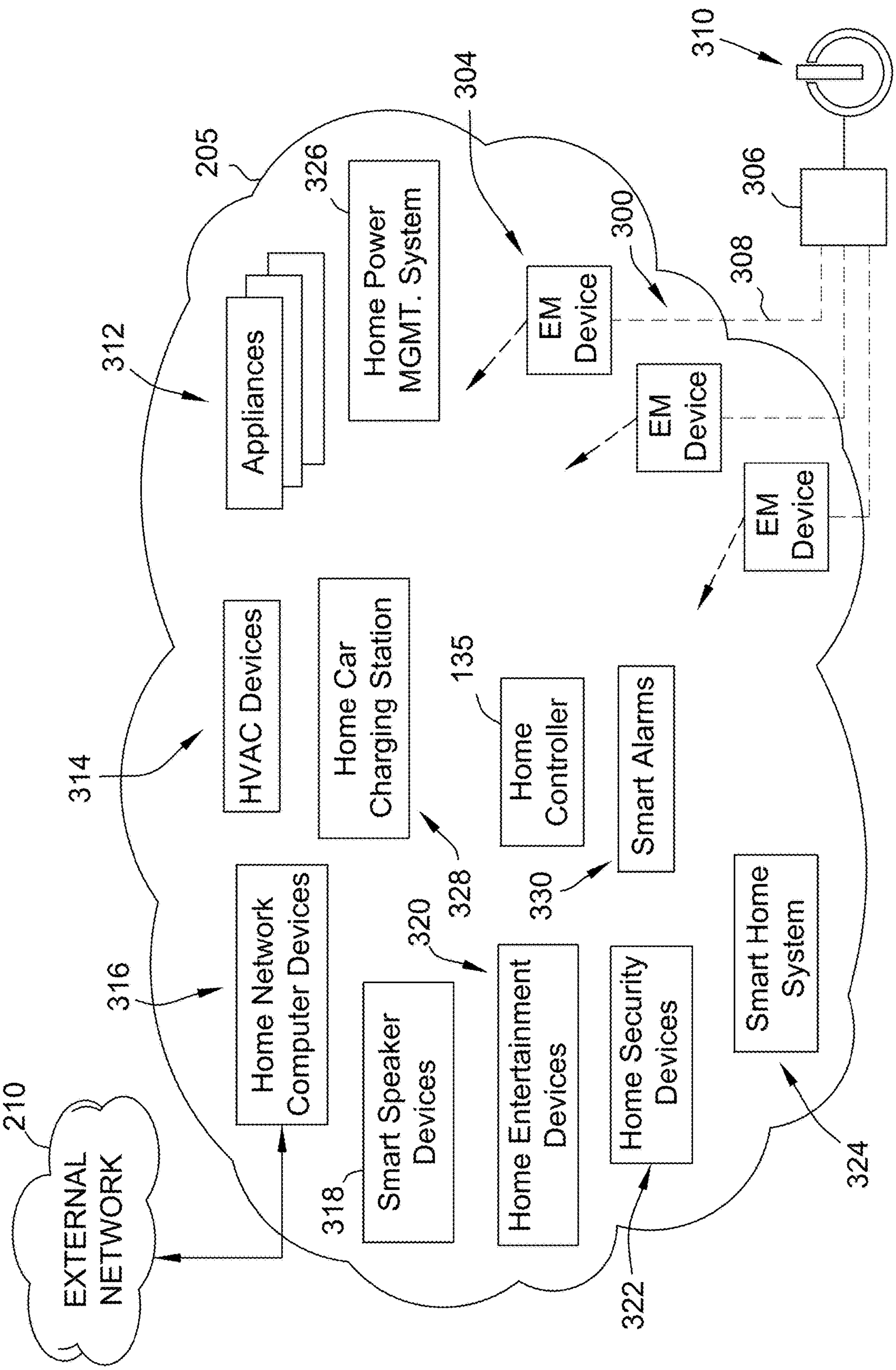

205
210
EXTERNAL NETWORK
312
Appliances
326
Home Power MGMT. System
304
EM Device
300
EM Device
EM Device
308
306
310
314
HVAC Devices
Home Car Charging Station
328
135
Home Controller
Smart Alarms
330
316
Home Network Computer Devices
318
Smart Speaker Devices
320
Home Entertainment Devices
322
Home Security Devices
324
Smart Home System

FIG. 3

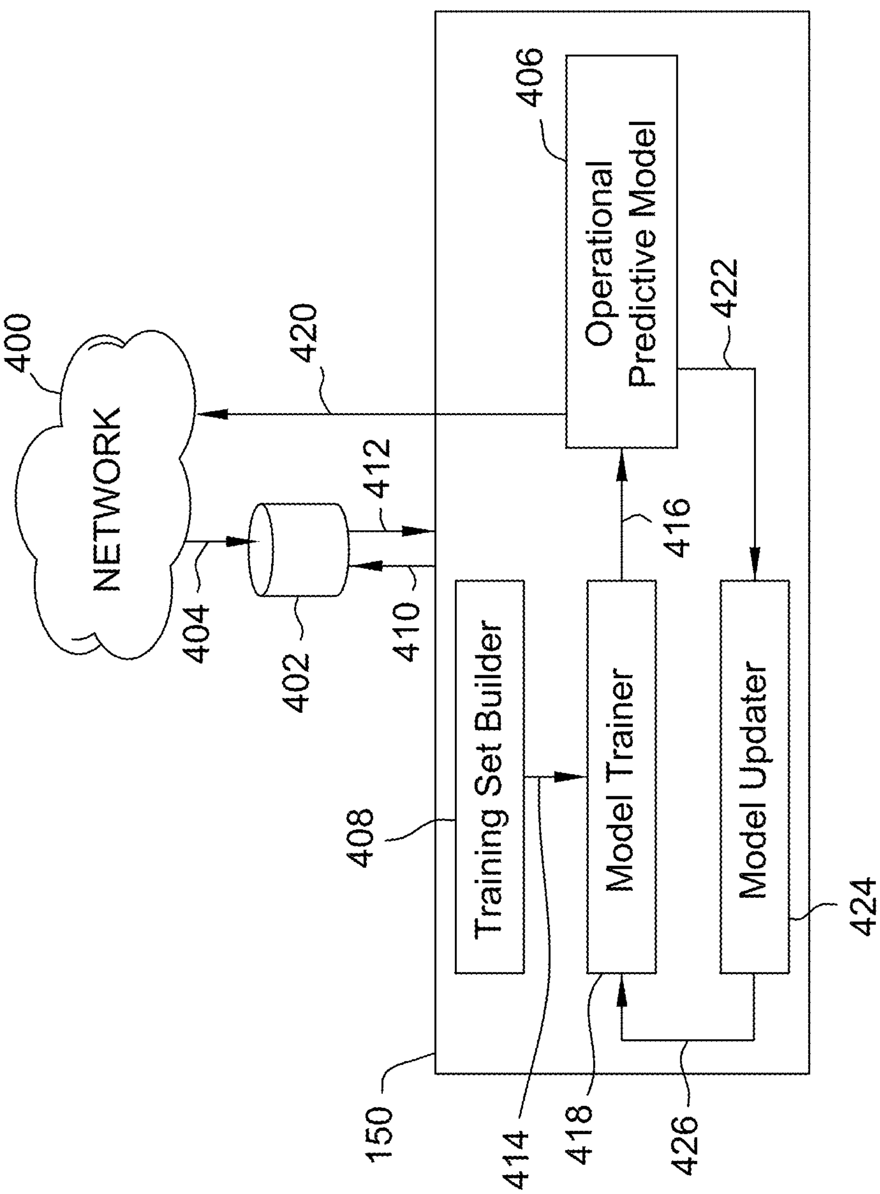

400
NETWORK
404
402
410
412
420
406
Operational Predictive Model
422
416
408
Training Set Builder
414
Model Trainer
418
Model Updater
424
426
150

FIG. 4

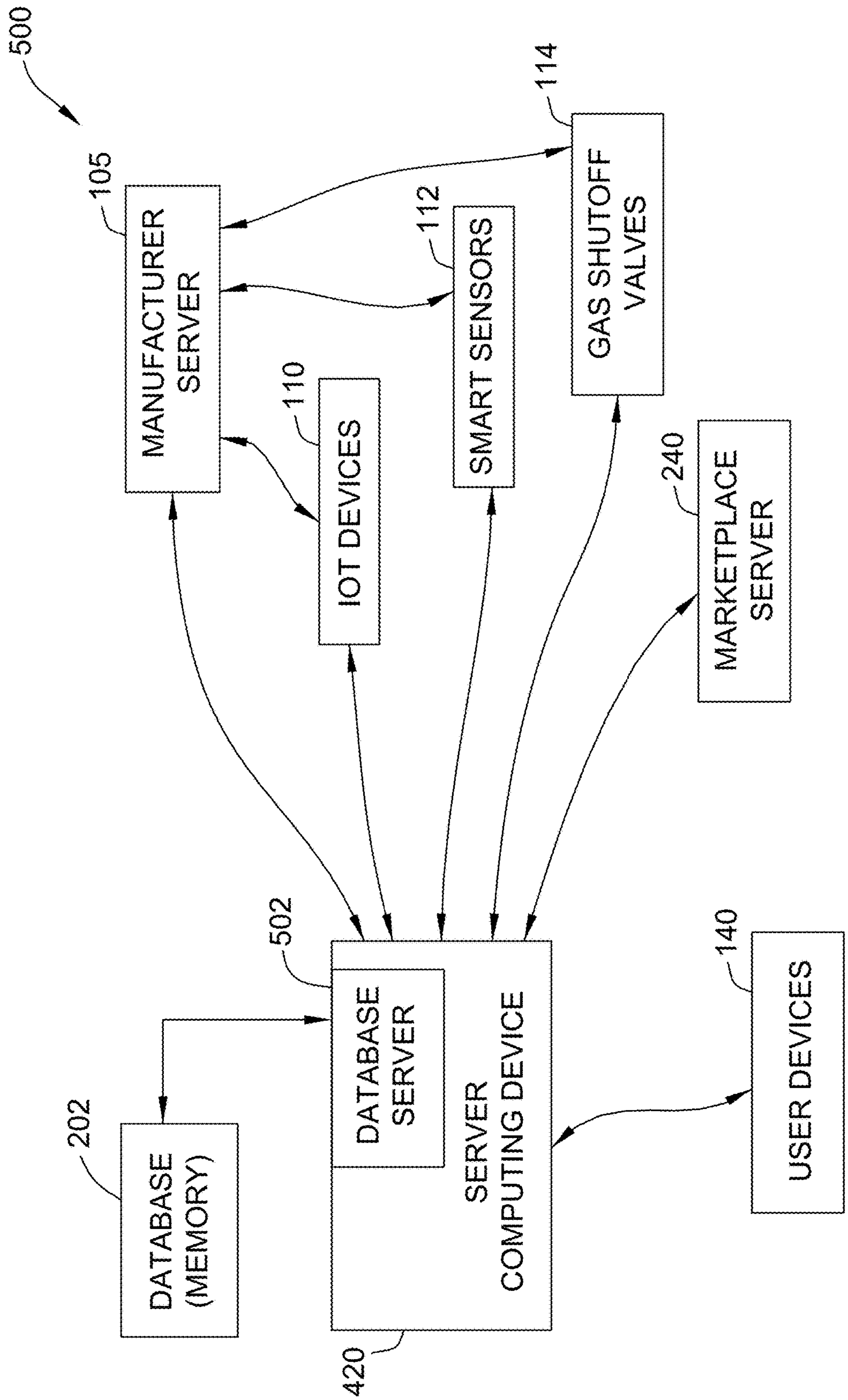

500
105
MANUFACTURER SERVER
110
IOT DEVICES
112
SMART SENSORS
114
GAS SHUTOFF VALVES
240
MARKETPLACE SERVER
502
DATABASE SERVER
SERVER COMPUTING DEVICE
420
202
DATABASE (MEMORY)
140
USER DEVICES

FIG. 5

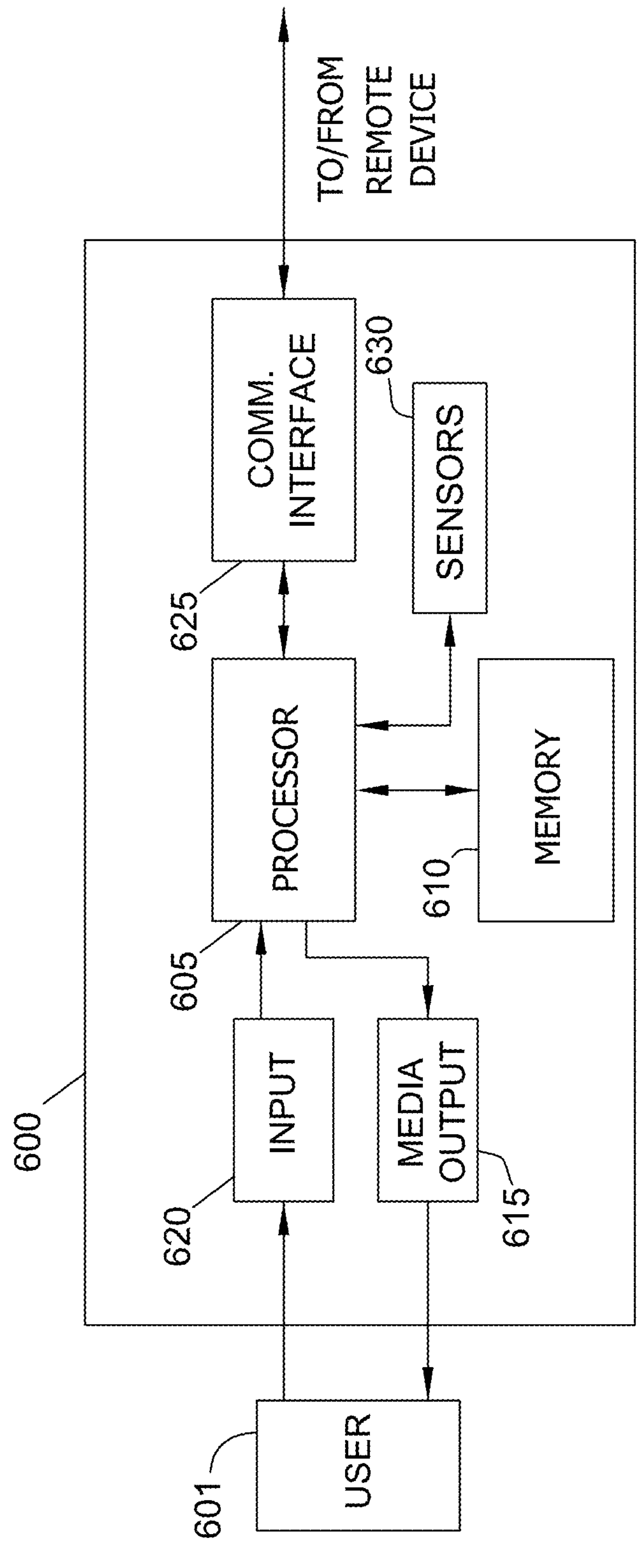

TO/FROM
REMOTE
DEVICE
COMM.
INTERFACE
625
630
SENSORS
PROCESSOR
MEMORY
610
605
600
INPUT
MEDIA
OUTPUT
620
615
USER
601

FIG. 6

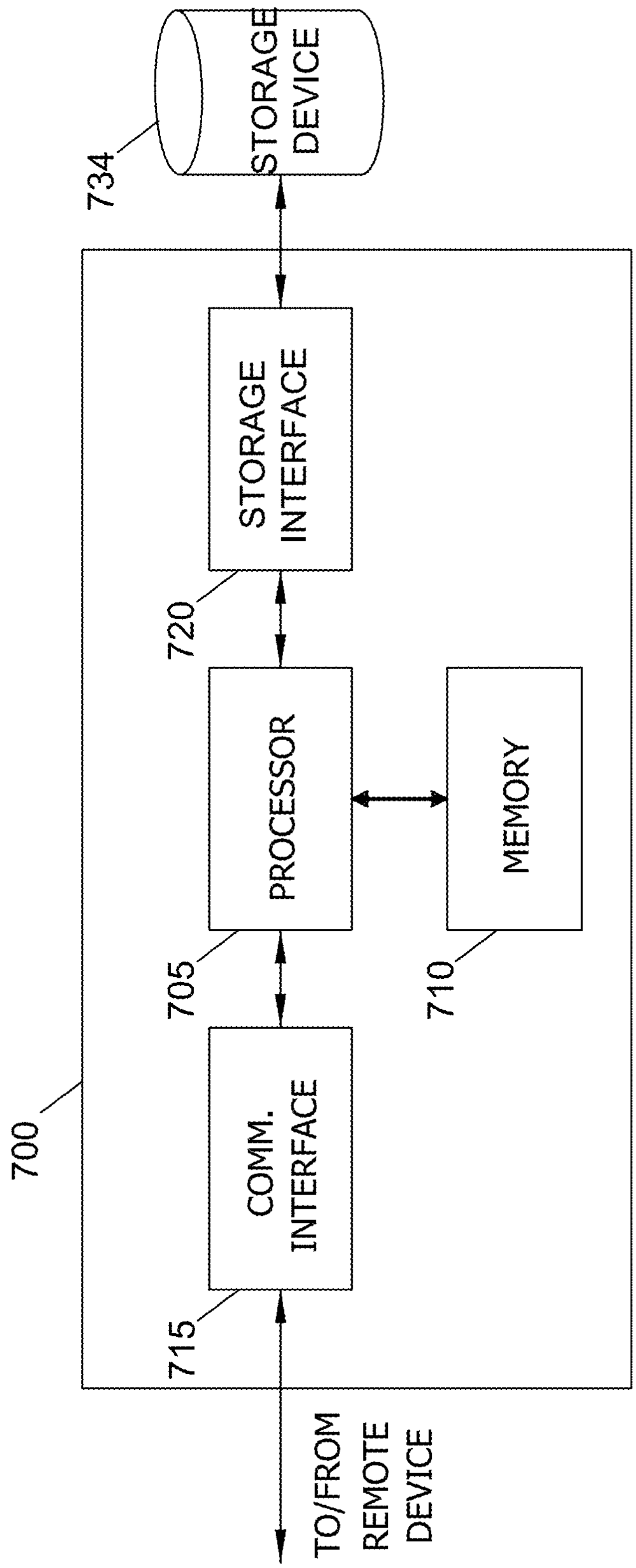

STORAGE DEVICE
734
700
STORAGE INTERFACE
720
PROCESSOR
705
MEMORY
710
COMM. INTERFACE
715
TO/FROM REMOTE DEVICE

FIG. 7

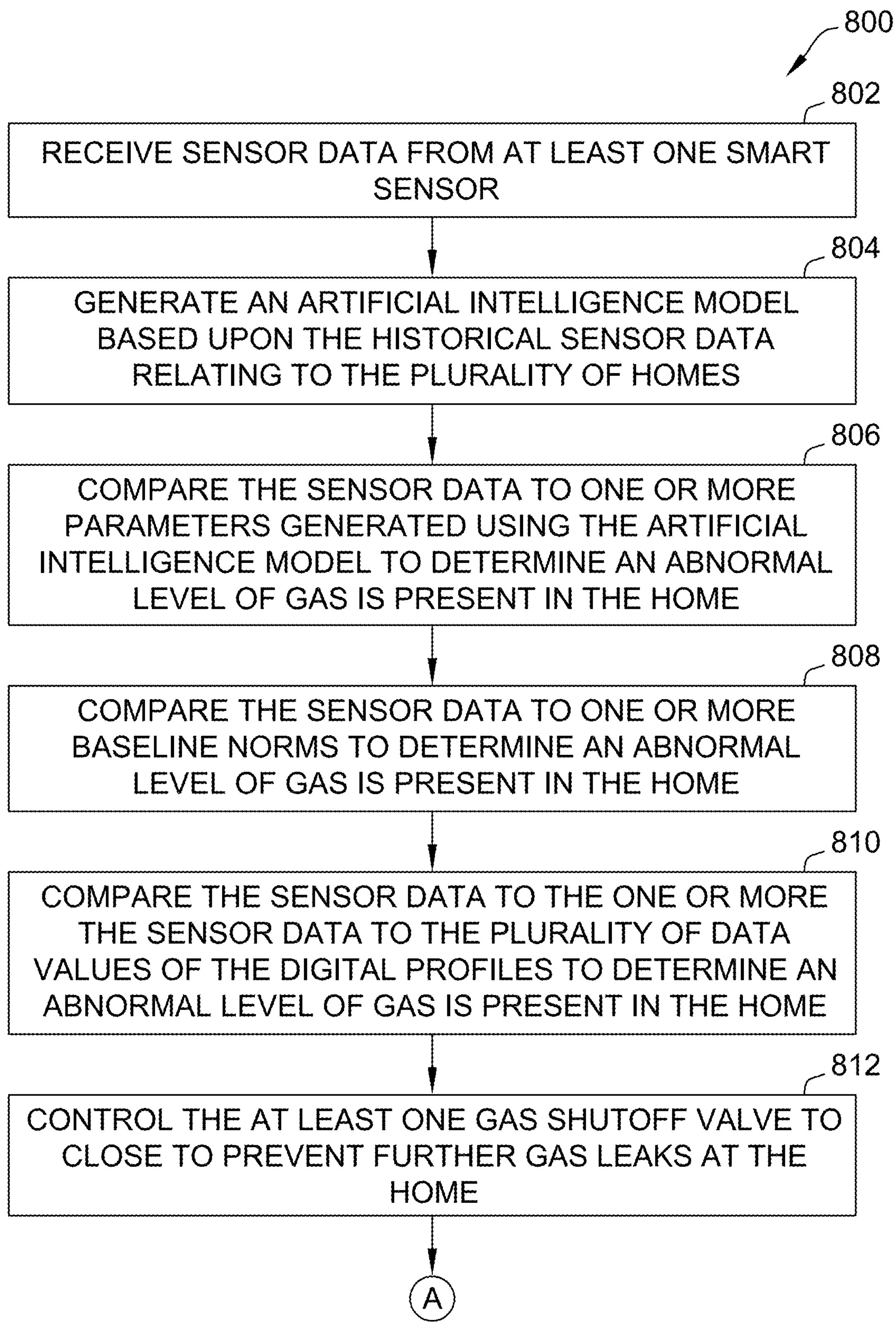

800
802
RECEIVE SENSOR DATA FROM AT LEAST ONE SMART SENSOR
804
GENERATE AN ARTIFICIAL INTELLIGENCE MODEL BASED UPON THE HISTORICAL SENSOR DATA RELATING TO THE PLURALITY OF HOMES
806
COMPARE THE SENSOR DATA TO ONE OR MORE PARAMETERS GENERATED USING THE ARTIFICIAL INTELLIGENCE MODEL TO DETERMINE AN ABNORMAL LEVEL OF GAS IS PRESENT IN THE HOME
808
COMPARE THE SENSOR DATA TO ONE OR MORE BASELINE NORMS TO DETERMINE AN ABNORMAL LEVEL OF GAS IS PRESENT IN THE HOME
810
COMPARE THE SENSOR DATA TO THE ONE OR MORE THE SENSOR DATA TO THE PLURALITY OF DATA VALUES OF THE DIGITAL PROFILES TO DETERMINE AN ABNORMAL LEVEL OF GAS IS PRESENT IN THE HOME
812
CONTROL THE AT LEAST ONE GAS SHUTOFF VALVE TO CLOSE TO PREVENT FURTHER GAS LEAKS AT THE HOME
A

FIG. 8A

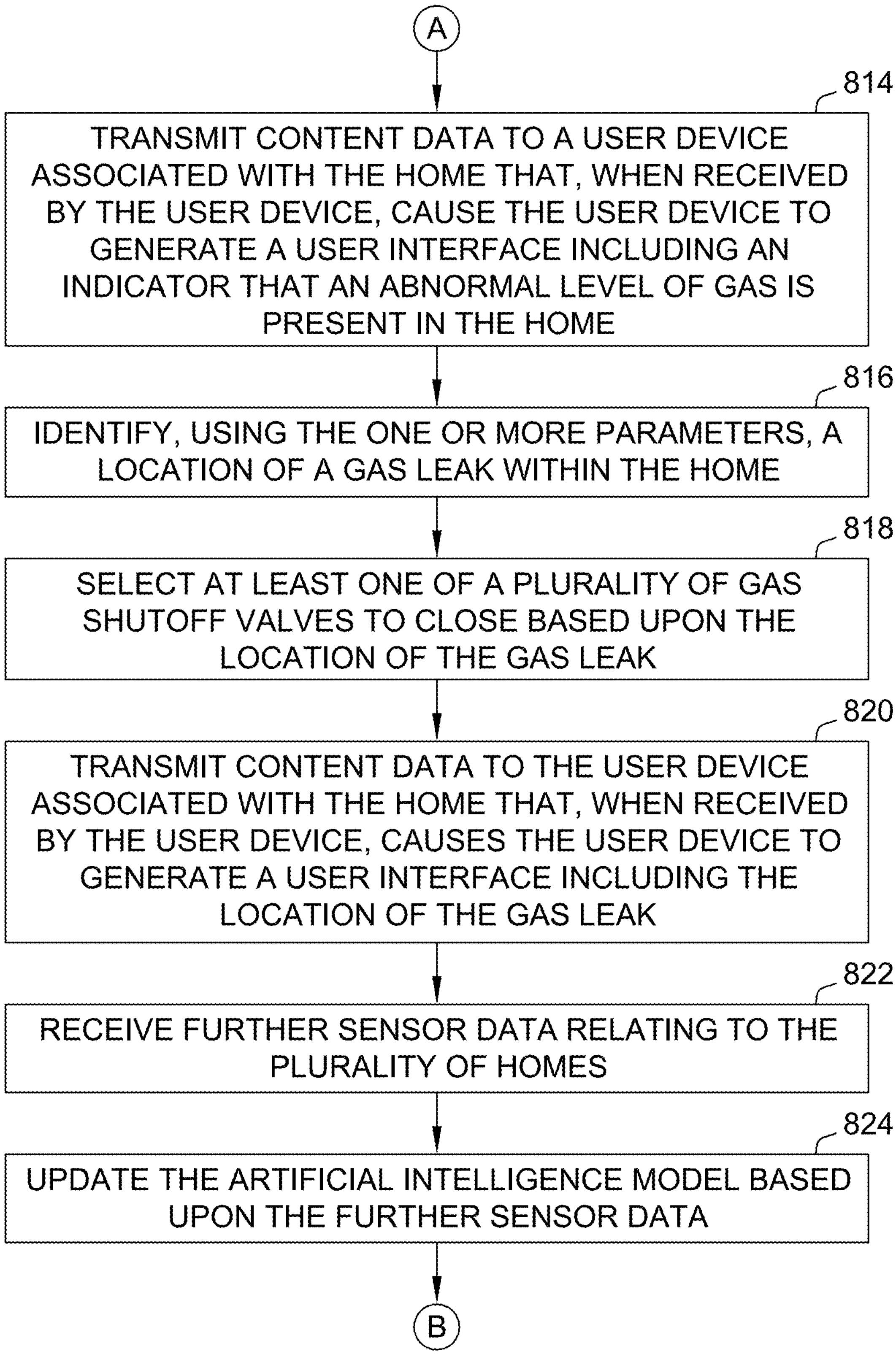

A
814
TRANSMIT CONTENT DATA TO A USER DEVICE ASSOCIATED WITH THE HOME THAT, WHEN RECEIVED BY THE USER DEVICE, CAUSE THE USER DEVICE TO GENERATE A USER INTERFACE INCLUDING AN INDICATOR THAT AN ABNORMAL LEVEL OF GAS IS PRESENT IN THE HOME
816
IDENTIFY, USING THE ONE OR MORE PARAMETERS, A LOCATION OF A GAS LEAK WITHIN THE HOME
818
SELECT AT LEAST ONE OF A PLURALITY OF GAS SHUTOFF VALVES TO CLOSE BASED UPON THE LOCATION OF THE GAS LEAK
820
TRANSMIT CONTENT DATA TO THE USER DEVICE ASSOCIATED WITH THE HOME THAT, WHEN RECEIVED BY THE USER DEVICE, CAUSES THE USER DEVICE TO GENERATE A USER INTERFACE INCLUDING THE LOCATION OF THE GAS LEAK
822
RECEIVE FURTHER SENSOR DATA RELATING TO THE PLURALITY OF HOMES
824
UPDATE THE ARTIFICIAL INTELLIGENCE MODEL BASED UPON THE FURTHER SENSOR DATA
B

FIG. 8B

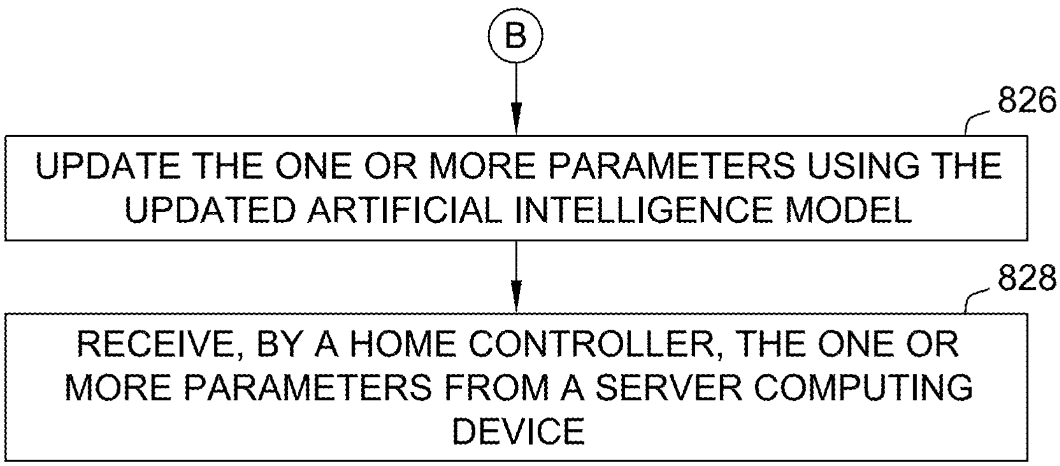

B
826
UPDATE THE ONE OR MORE PARAMETERS USING THE UPDATED ARTIFICIAL INTELLIGENCE MODEL
828
RECEIVE, BY A HOME CONTROLLER, THE ONE OR MORE PARAMETERS FROM A SERVER COMPUTING DEVICE

FIG. 8C

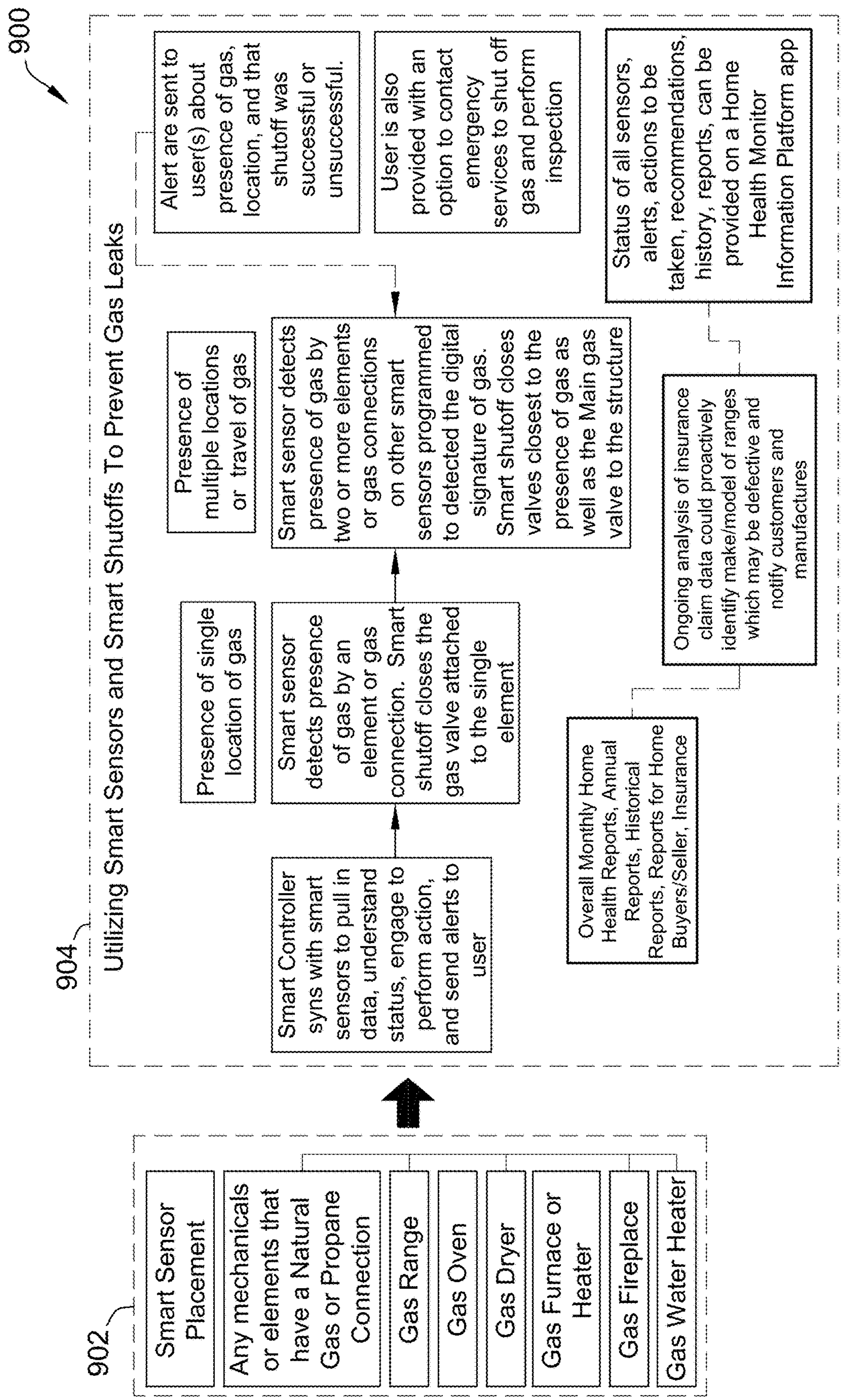

900
902
Smart Sensor Placement
Any mechanicals or elements that have a Natural Gas or Propane Connection
Gas Range
Gas Oven
Gas Dryer
Gas Furnace or Heater
Gas Fireplace
Gas Water Heater
904
Utilizing Smart Sensors and Smart Shutoffs To Prevent Gas Leaks
Smart Controller syns with smart sensors to pull in data, understand status, engage to perform action, and send alerts to user
Presence of single location of gas
Smart sensor detects presence of gas by an element or gas connection. Smart shutoff closes the gas valve attached to the single element
Presence of multiple locations or travel of gas
Smart sensor detects presence of gas by two or more elements or gas connections on other smart sensors programmed to detected the digital signature of gas. Smart shutoff closes valves closest to the presence of gas as well as the Main gas valve to the structure
Alert are sent to user(s) about presence of gas, location, and that shutoff was successful or unsuccessful.
User is also provided with an option to contact emergency services to shut off gas and perform inspection
Overall Monthly Home Health Reports, Annual Reports, Historical Reports, Reports for Home Buyers/Seller, Insurance
Ongoing analysis of insurance claim data could proactively identify make/model of ranges which may be defective and notify customers and manufactures
Status of all sensors, alerts, actions to be taken, recommendations, history, reports, can be provided on a Home Health Monitor Information Platform app

FIG. 9

SYSTEMS AND METHODS FOR GAS MONITORING AND GAS LEAK PREVENTION

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority of U.S. Provisional Patent Application No. 63/560,477, filed Mar. 1, 2024, and entitled "SYSTEMS AND METHODS FOR GAS MONITORING AND GAS LEAK PREVENTION," the contents and disclosures of which are hereby incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The field of the disclosure relates generally to gas leak monitoring, and more specifically, to a smart home computing system for detecting and preventing gas leaks in a home using smart sensors and smart shutoff valves.

BACKGROUND

[0003] Issues within a home or other building may not always be easy to detect. These issues may grow worse if not detected and addressed early. For example, different gasses used as fuel to heat, cool, or otherwise power homes, such as natural gas and propane, may be combustible and pose a danger of fire, injury, or health problems when a gas leak occurs. However, gas leaks may often be not perceptible to a person, and when a presence of gas is detected, it may be difficult to determine where the gas leak occurred within a home or building. A system capable of identifying gas leaks and their locations and taking remedial action before the gas leaks poses a significant risk of harm or damage is therefore desirable. Conventional techniques may include additional inefficiencies, encumbrances, ineffectiveness, and/or other drawbacks as well.

BRIEF DESCRIPTION

[0004] The present embodiments may relate to, inter alia, computer systems and computer-based methods that retrieve sensor data from smart sensors located within homes. This sensor data may be compared to parameters that are generated using a machine learning and/or AI model that predict when an abnormal level of gas (e.g., natural gas or propane) is present in a home. This model may be trained using historical data retrieved from a large number of homes. Based upon this comparison, the system may control gas shutoff valves within the home to prevent further leaking of gas, and may generate alerts, recommendations, and/or reports that may be presented to associated homeowners and/or other users. The use of the machine learning and/or AI modeling (and/or other AI and/or machine learning techniques) may be available in various mediums such as a computer and/or mobile application, chat screens, web page, voice interaction with a voice chat-capable connected home device, voice bots or chat bots, ChatGPT bots, and/or social media messaging. The system may include less, or alternate functionality, including that discussed elsewhere herein.

[0005] In one aspect, a computer system for monitoring levels of gases in a home may be provided. The system may include one or more local or remote processors, servers, sensors, transceivers, mobile devices, wearables, smart watches, smart contact lenses, voice bots, chat bots, ChatGPT bots, augmented reality glasses, virtual reality headsets, mixed or extended reality headsets or glasses, and other electronic or electrical components, which may be in wired or wireless communication with one another. For example, in one instance, the computer system may be programmed to: (a) receive sensor data from at least one smart sensor; (b) compare the sensor data to one or more parameters generated using an artificial intelligence model to determine that an abnormal level of gas is present in the home, wherein the artificial intelligence model is trained based upon historical sensor data relating to a plurality of homes; and/or (c) in response to determining that an abnormal level of gas is present in the home, control at least one gas shutoff valve to close to prevent further gas leaks at the home. The computer system may include additional, less, or alternate functionality, including that discussed elsewhere herein.

[0006] In yet another aspect, a computer-implemented method for monitoring of a home may be provided. The computer-implemented method may be implemented via one or more local or remote processors, servers, sensors, transceivers, mobile devices, wearables, smart watches, smart contact lenses, voice bots, chat bots, ChatGPT bots, augmented reality glasses, virtual reality headsets, mixed or extended reality headsets or glasses, and other electronic or electrical components, which may be in wired or wireless communication with one another. For example, in one instance, the computer-implemented method may be performed by a computing device including at least one processor and at least one memory device. The method may include, via the at least one processor: (a) receiving sensor data from at least one smart sensor; (b) comparing the sensor data to one or more parameters generated using an artificial intelligence model to determine an abnormal level of gas is present in the home, wherein the artificial intelligence model is trained based upon historical sensor data relating to a plurality of homes; and/or (c) in response to determining that an abnormal level of gas is present in the home, controlling at least one gas shutoff valve to close to prevent further gas leaks at the home. The method may have additional, less, or alternate actions, including that discussed elsewhere herein.

[0007] In still another aspect, at least one non-transitory computer-readable medium having computer-executable instructions embodied thereon may be provided. The computer-executable instructions may be executed using one or more local or remote processors, servers, sensors, transceivers, mobile devices, wearables, smart watches, smart contact lenses, voice bots, chat bots, ChatGPT bots, augmented reality glasses, virtual reality headsets, mixed or extended reality headsets or glasses, and other electronic or electrical components, which may be in wired or wireless communication with one another. For example, in one instance, when executed by at least one processor, the computer-executable instructions cause the at least one processor a to: (a) receive sensor data from at least one smart sensor; (b) compare the sensor data to one or more parameters generated using an artificial intelligence model to determine an abnormal level of gas is present in the home, wherein the artificial intelligence model is trained based upon historical sensor data relating to a plurality of homes; and/or (c) in response to determining that an abnormal level of gas is present in the home, control at least one gas shutoff valve to close to prevent further gas leaks at the home. The computer-readable medium may have instructions that direct additional, less, or alternate functionality, including that discussed elsewhere herein.

[0008] Advantages will become more apparent to those skilled in the art from the following description of the preferred embodiments which have been shown and described by way of illustration. As will be realized, the present embodiments may be capable of other and different embodiments, and their details are capable of modification in various respects. Accordingly, the drawings and description are to be regarded as illustrative in nature and not as restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The figures described below depict various aspects of the systems and methods disclosed therein. It should be understood that each figure depicts an embodiment of a particular aspect of the disclosed systems and methods, and that each of the figures is intended to accord with a possible embodiment thereof. Further, wherever possible, the following description refers to the reference numerals included in the following figures, in which features depicted in multiple figures are designated with consistent reference numerals.

[0010] There are shown in the drawings arrangements which are presently discussed, it being understood, however, that the present embodiments are not limited to the precise arrangements and are instrumentalities shown, wherein:

[0011] FIG. **1** illustrates an exemplary computer system for home monitoring in accordance with the present disclosure.

[0012] FIG. **2** illustrates an expanded home monitoring, analysis, and marketplace system including the exemplary system of FIG. **1**.

[0013] FIG. **3** illustrates exemplary source devices that may be used with the systems shown in FIGS. **1** and **2**.

[0014] FIG. **4** illustrates an exemplary server computing device for use in the systems shown in FIGS. **1** and **2**.

[0015] FIG. **5** illustrates an exemplary computer system for implementing the systems shown in FIGS. **1** and **2** performing the computer-implemented method shown in FIG. **8**.

[0016] FIG. **6** depicts an exemplary configuration of a client computer device in accordance with one embodiment of the present disclosure.

[0017] FIG. **7** depicts an exemplary configuration of a server computing device in accordance with one embodiment of the present disclosure.

[0018] FIG. **8**A depicts a flowchart of an exemplary computer-implemented method for gas monitoring in a home using the systems shown in FIGS. **1** and **2**.

[0019] FIG. **8**B is a continuation of the flowchart shown in FIG. **8**A.

[0020] FIG. **8**C is a continuation of the flowchart shown in FIGS. **8**A and **8**B.

[0021] FIG. **9** is a diagram illustrating an exemplary data analysis process for gas monitoring in a home that may be performed by the systems shown in FIGS. **1** and **2**.

[0022] The figures depict preferred embodiments for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the systems and methods illustrated herein may be employed without departing from the principles of the invention described herein.

DETAILED DESCRIPTION

[0023] The present embodiments may relate to, inter alia, computer systems and computer-based methods that retrieve sensor data from smart sensors located within homes. This sensor data may be compared to parameters that are generated using a machine learning and/or AI model that predict when an abnormal level of gas (e.g., natural gas or propane) is present in a home. This model may be trained using historical data retrieved from a large number of homes. Based upon this comparison, the system may control gas shutoff valves within the home to prevent further leaking of the gas, control ventilation systems to help quickly dissipate the gas present in the home, and may generate alerts, recommendations, and/or reports that may be presented to associated homeowners and/or other users.

[0024] In the exemplary embodiment, the system may include a home controller that communicates with at least one smart sensor and at least one gas shutoff valve disposed within a home. The smart sensors may include free-standing devices and/or be incorporated into other devices (e.g., smart appliances), and may include different types of sensors configured to generate corresponding types of sensor data. For example, these sensors may include gas sensors of other sensors that generate data from which an amount of gas present and/or a presence and location of a gas leak may be inferred. These gas sensors may be positioned in or near gas appliances or at other strategic locations within the home.

[0025] As described in further detail below, combinations of different types of sensor data taken from different locations within the home may be compared to certain predefined parameters to identify, for example, if an abnormal level of gas is present in the home, if a gas leak is present, and/or locations of abnormal levels of gas and/or gas leaks. The gas shutoff valves may be installed on gas lines and/or integrated into certain gas appliances, and may be controlled by the home controller, for example, to stop a flow of gas to areas determined to be a likely location of a gas leak. In some embodiment, the system may also control ventilation systems (e.g., HVAC, exhaust fans, and/or other fans or controllable vents) in the area of the gas leak to quickly dissipate the buildup of gas.

[0026] In the exemplary embodiment, the system may be configured to generate parameters that identify when an abnormal level of gas is present in the home, if a gas leak is present, and/or locations of abnormal levels of gas and/or gas leaks within the home by querying a machine learning and/or AI model, such as a large language trained (LLM) generative AI model. This AI model may be trained using historical data relating to historical sensor data.

[0027] The historical data may include different types of historical sensor data and historical conditions and/or outcomes associated with the historical sensor data. For example, the AI model may define a correlation between historical instances of abnormal or unsafe levels of gas, gas leaks, and their locations and sensor data or other contextual data that has historically occurred in connection with these issues. Other data considered by the model may include structural details and/or geographic information about the home, household habits or practices of those living in the home (e.g., usage of gas appliances), and/or any of the other factors that could be indicative of these issues occurring within a home. This data may be generated by sensors and/or include self-reported data.

[0028] In some embodiments, the AI model may be built and/or trained by one or more server computing devices, and parameters output by the AI model and/or the AI model itself may be transmitted to various home controllers for analyzing sensor data of respective homes. Alternatively, in some embodiments, the AI model may be trained by local home controllers. In some embodiments, the system may (e.g., continually) receive further sensor data, based upon which the system may continually and/or periodically update the AI model. As the AI model is updated or re-trained, the AI model may be used to generate new parameters for analyzing sensor data, which may be transmitted to local home controllers.

[0029] In the exemplary embodiment, the system may be configured to compare the sensor data received from smart sensors of a home to one or more of the parameters generated using an artificial intelligence model to identify an abnormal level of gas is present in the home, if a gas leak is present, and/or locations of abnormal levels of gas and/or gas leaks within the home. For example, in some embodiments, the parameters may include one or more baseline norms, or expected values or ranges for certain types of sensor data (e.g., gas levels in the air and/or other related types of data), to which the system may compare the sensor data to identify an abnormal level of gas is present in the home, if a gas leak is present, and/or locations of abnormal levels of gas and/or gas leaks within the home. For instance, some amount of gas may be normal in certain locations base upon, for example, whether the location is near a gas appliance, whether the area is ventilated, whether the location is in a home, outdoors, or another type of building, etc., but an amount of gas in excess of this normal amount may be a cause for attention and/or taking remedial measures. These baselines norms may be generated by the AI model based upon historical sensor data.

[0030] In some embodiments, certain alert conditions may require a combination of different baseline norms, or that certain length or severity thresholds, or combinations thereof, be met. For example, if a relatively high gas level compared to the baseline norm is detected, but only for a relatively short period of time that does not periodically reoccur, the system may record and/or generate alerts for the high gas level but may not necessarily take other measures, such as closing a gas shutoff valve. However, if a high level compared to the baseline norm is detected for a threshold period of time and/or if the high level occurs periodically (e.g., whenever a certain gas appliance is first activated), the system may determine there is a higher risk and implement further measures, such as closing the gas shutoff valve.

[0031] In some embodiments, the parameters may include one or more digital profiles including a plurality of data values corresponding to different types of sensor data, and the system may compare the sensor data to the plurality of data values of the digital profiles to identify the at least one alert condition. For example, a digital profile may be associated with a presence of abnormal levels of gas, a gas leak, or a certain location of abnormal levels of gas or a gas leak within a home. If the sensor data includes readings that match this digital profile, the system may determine that there is a presence of abnormal levels of gas, a gas leak, or a certain location of abnormal levels of gas or a gas leak within a home.

[0032] In the exemplary embodiment, the system may be configured to control one or more shutoff valves within the home to close in response to determining that an abnormal level of gas is present in the home. In some embodiments, in which a specific location of abnormal levels of gas and/or a gas leak within the home is determined, the system may select certain gas shutoff valves to close based upon the location. For example, if a certain appliance is determined to be leaking, the system may determine to close a gas shutoff valve specific to that appliance. If abnormal levels of gas and/or leaks are detected in multiple locations, or if gas levels remain present or continue to rise after an initial local shutoff, a main gas shutoff valve to the house may be closed. Examples of fixtures, appliances, or locations that may include gas shutoff valves include a main supply valve (e.g., at the point of entry or gas meter, a gas range, a gas oven, a furnace, a water heater, a fireplace, a gas heater, a gas dryer, an outdoor grill, an external gas portal for lawn or other equipment, or any other gas appliance and/or access point. In some embodiments, certain appliances (e.g., smart appliances connected to the home controller) may include integrated gas shutoff valves. Other appliances may have a supply of gas controlled by a shutoff valve installed on a gas line leading to the appliance.

[0033] Similarly, in some embodiments, in response to determining that an abnormal level of gas is present in the home, the system may identify one or more ventilation systems (e.g., HVAC, exhaust fans, and/or other fans or controllable vents) capable of dispersing buildups of gas at the location of the gas leak or abnormally high levels of gas and activating these identified ventilation systems.

[0034] In some embodiments, the system may be configured to transmit data (referred to herein as “content data”) to a user device associated with the home that, when received by the user device, causes the user device to generate a user interface including any identified alert conditions as to a presence of abnormal levels of gas, a gas leak, or a certain location of abnormal levels of gas or a gas leak within the home and the status of any gas shutoff valves or ventilation systems configured to dissipate gas (e.g., which gas shutoff valves have been closed in response to the detection of abnormal levels, whether ventilation systems were activated, and/or whether such operations were successful). For example, the user device may be a mobile device or other computing device capable of executing a mobile application (“app”) that causes the user interface to be displayed on the user devices.

[0035] In certain embodiments, alert notifications may be displayed automatically whenever abnormal levels of gas are detected and/or gas shutoff valves are closed, and nofictions may also be automatically transmitted to emergency personnel, gas utility companies, and/or any other relevant parties. The app may include user-selectable options, for example, to contact emergency services or to have gas shut off and an inspection performed by a gas utility company. In addition to any identified alert conditions, the user interface may also display other data and recommendations. For example, the user interface to display numbers, indicators, or graphics representing at least some of the received sensor data (e.g., current or historical gas levels), which may illustrate outliers in senor data, locations of detected issues, graphs of norms and current levels, trends over time, comparisons to thresholds considered safe or normal, and comparisons to other homes (e.g., averages in a neighborhood and/or geographic area of the homes). In some such embodiments, this data may be continuously or periodically updated to reflect real time conditions within the home. The user interface may enable the user to close gas shutoff valves and/or activate ventilation systems remotely from the user device.

[0036] In some embodiments, the system may be configured to generate at least one recommendation based upon the identified alert condition and cause the user interface to display the generated recommendation. For example, the system may identify a certain gas appliance as being a source of a leak and recommend the appliance be replaced, and/or may identify a location of a gas line within the home that is leaking and recommend that emergency and/or repair services may be contacted. In some embodiments, if the system determines a risk associated with a particular leak is sufficiently high even if measures such as gas shutoff are implemented, the system may automatically contact emergency, gas utility, or other appropriate services. Additionally or alternatively, certain recommendations nay be generated prophylactically even if a gas leak is not detected. For example, if a certain model of appliance that is determined to be prone to gas leaks is present in the home, the system may generate a recommendation that this appliance be replaced.

[0037] In certain embodiments, AI or chatbots may be used to generate recommendations. An output of the AI or chatbot may include computer executable instructions for generating a user interface (e.g., within the app and/or web page) to present the recommendations. The use of the generative AI model may be available in various mediums such as a computer and/or mobile application, chat screens, web page, voice interaction with a voice chat-capable connected home device, voice bots or chat bots, ChatGPT bots, and/or social media messaging. The system may generate reports (e.g., on demand or periodically) indicating any issues that were identified and remedies taken, which may be provided to potential buyers of the home or other parties such as gas utility companies and/or insurers.

[0038] In various embodiments, the system (e.g., using an AI model) may search for products, stores, and/or services relating to the recommendations and, in some such embodiments, provide a link (e.g., a hyperlink) to the recommended products, stores, and/or services to the user via a computing device (e.g., via a mobile application, web page, and/or email). For example, the links may relate to products and/or maintenance services that may address alert conditions and/or other issues identified in the home (e.g., replacing leaking appliances and/or contact acting repair and/or installation services). The system may use geolocation to determine appropriate products, stores, home services, and/or emergency services located near, at, or around the vicinity of the user's geolocation, which the homeowner may obtain in order to implement recommendations provided by the system.

[0039] In some embodiments, the recommendations may include a schedule of actions and/or a curated plan and set of reminders for when certain actions should be performed. Additionally or alternatively, the recommendations may include a list actions to address potential issues identified in the home. The recommendations may include a timeline and/or a recommended order for performing these actions. The recommendation list may be presented through a user interface that enables the homeowner to select and/or click on listed actions to automatically purchase and/or schedule products and/or services associated with the actions. For example, if an appliance that leaks gas is identified, the system may provide a link to a repair service and/or to purchase a replacement appliance.

[0040] The system may further provide additional information relating to the recommended actions, such as advantages of the particular actions, costs, potential cost savings (e.g., reducing insurance and/or energy costs), when and how to perform the actions, and/or alternative options. In some embodiments, the recommendations may be generated in response to query by the homeowner and/or may be periodically generated automatically (e.g., as a monthly report to the homeowner).

[0041] In some embodiments, the system may be communicably coupled to a communication network and/or a financial services provider. The system may receive insurance information from the financial services provider. The system may connect an insurance policy of the homeowner to the generated recommendations, in which the application may display potential changes to the homeowner's insurance policy based upon, for example, installation of smart sensors in the home, a presence of certain conditions, and/or implementation of recommendations that fix and/or otherwise mitigate these conditions. The system may prioritize the recommendations based upon potential changes to a customer's insurance policy or claims information submitted by other customers living in an area geolocated near the homeowner.

[0042] In some embodiments, the system may include a risk evaluation engine that may evaluate sensor data associated with a home to evaluate various risks associated with the home. For example, certain data patterns or profiles may correlate to a certain level of risk of damage occurring to the home. The system may use numerous data points to evaluate such risks to a residential property and may compute a composite risk score and/or various focused risk scores for the property. The risk score (e.g., or likelihood of damage score) may be a numeric value and/or a category (e.g., excellent, good, fair, and poor). Such risk scores may be used, for example by an insurance provider, to evaluate insurability of the property and its assets, to price insurance policy options for the property, or to provide policy discounts and verify compliance for risk mitigating changes, actions, or behaviors. Further, such risk scores may be used to determine to recommend actions (e.g., maintenance) be taken.

[0043] In certain embodiments, the system may generate a risk score for different categories of risk, such as property risk, fire protection, and safety, which may be presented individually within the user interface with related recommendations. For example, the fire protection rating may be displayed along with fire-protection related recommendations, such as recommendation relating to actions that may result in a reduction of fire risk if implemented (e.g., addressing alert conditions identified by the system that relate to fire safety).

[0044] While various examples provided herein describe application of the system to various aspects of home appliances and other home systems, the systems and methods described herein may also be used for performing other analysis, such as vehicles, businesses, municipal locations, and/or other locations and/or items.

Exemplary Computer System for Gas Monitoring in a Home

[0045] FIG. **1** illustrates an exemplary computer system **100** for monitoring and analyzing homes, in accordance with at least one embodiment of this disclosure. System **100** illustrates monitoring and other devices to receive, analyze, and report the data collected about the home.

[0046] In the exemplary embodiment, a manufacturer server **105** provides one or more IoT devices **110**, also known as IoT devices **110**, smart sensors **112**, and/or gas shutoff valves **114**. These IoT devices **110** and/or smart sensors **112** may be positioned in or around a home **130**. IoT devices **110** may include, but are not limited to IoT cameras **115**, IoT thermostats **120**, IoT door locks **125**, and/or any other internet connected device, including, but not limited to, user devices **140**, which may be mobile devices, laptops, and/or a mobile phones, one or more voice or chat bots, a computer device, including, but not limited to, a desktop computer and/or a router, and/or a home controller **135**.

[0047] In at least one embodiment, the home controller **135** is in wired or wireless communication the one or more IoT devices **110**, smart sensors **112**, and/or gas shutoff valves **114** in home **130**. In some embodiments, the home controller **135** may be a router or Wi-Fi providing device in the home **130**. In other embodiments, the home controller **135** is a smart home controller that controls one or more of IoT devices **110**, smart sensors **112**, and/or gas shutoff valves **114** and may provide communication between the user and the individual IoT devices **110**, smart sensors **112**, and/or gas shutoff valves **114**.

[0048] In certain embodiments, each IoT devices **110**, smart sensors **112**, and/or gas shutoff valves **114** may collect data about the home either directly or indirectly. For example, a smart light bulb may report when the bulb is on and off, or a gas shutoff valve **114** may report whether it is open or closed.

[0049] In the at least one embodiment, many IoT devices **110**, smart sensors **112**, and/or gas shutoff valves **114** are in communication with one or more servers of the manufacturer server **105**. The manufacturer server **105** may provide additional services, such as remote activation. The manufacturer server **105** may also collect data observed by IoT device **110** and/or smart sensors **112**, including, but not limited to, sensor data and/or usage data about IoT device **110** and/or gas shutoff valves **114**.

[0050] In various embodiments, a server computing device **150** may be in communication with one or more of the IoT devices **110**, smart sensors **112**, gas shutoff valves **114**, the home controller **135**, and/or the manufacturer servers **105**. Server computing device **150** may collect data from IoT devices **110** and/or smart sensors **112** for use in identifying potential alert conditions (e.g., gas levels) or other issues in home **130** and generating corresponding scores and/or recommendations. Server computing device **150** may be in communication with one or more user devices **140** associated with respective homeowners, though which server computing device **150** may present alerts, sensor data, scores and/or recommendations, as described in further detail below.

[0051] In some embodiments, smart sensors **112** may include free-standing devices and/or be incorporated into other devices (e.g., IoT devices **110**), and may include different types of sensors configured to generate corresponding types of sensor data. For example, these sensors may include gas sensors of other sensors that generate data from which an amount of gas present and/or a presence and location of a gas leak can be inferred. These gas sensors may be positioned in or near gas appliances or at other strategic locations within home **130**.

[0052] As described in further detail below, combinations of different types of sensor data taken from different locations within the home may be compared to certain predefined parameters to identify, for example, if an abnormal level of gas is present in the home, if a gas leak is present, and/or locations of abnormal levels of gas and/or gas leaks. Gas shutoff valves **114** may be installed on gas lines and/or integrated into certain gas appliances, and may be controlled by home controller **135**, for example, to stop a flow of gas to areas determined to be a likely location of a gas leak. In some embodiment, home controller **135** and/or server computing device **150** may also control ventilation systems (e.g., HVAC, exhaust fans, and/or other fans or controllable vents) in the area of the gas leak to quickly dissipate the buildup of gas.

[0053] In the exemplary embodiment, server computing device **150** and/or home controller **135** may be configured to generate parameters that identify when an abnormal level of gas is present in the home, if a gas leak is present, and/or locations of abnormal levels of gas and/or gas leaks within the home by querying a machine learning and/or AI model, such as a large language trained (LLM) generative AI model. This AI model may be trained using historical data relating to historical sensor data.

[0054] The historical data may include different types of historical sensor data and historical conditions and/or outcomes associated with the historical sensor data. For example, the AI model may define a correlation between historical instances of abnormal or unsafe levels of gas, gas leaks, and their locations and sensor data or other contextual data that has historically occurred in connection with these issues. Other data considered by the model may include structural details and/or geographic information about home **130**, household habits or practices of those living in the home (e.g., usage of gas appliances), and/or any of the other factors that could be indicative of these issues occurring within a home. This data may be generated by sensors and/or include self-reported data.

[0055] In some embodiments, the AI model may be built and/or trained by server computing device **150**, and parameters output by the AI model and/or the AI model itself may be transmitted to home controller **135** for analyzing sensor data of respective homes. Alternatively, in some embodiments, the AI model may be trained by local home controllers **135**. The server computing device **150** and/or home controller **135** may (e.g., continually) receive further sensor data, based upon which server computing device **150** and/or home controller **135** may continually and/or periodically update the AI model. As the AI model is updated or re-trained, the AI model may be used to generate new parameters for analyzing sensor data, which may be transmitted to local home controllers **135**.

[0056] In the exemplary embodiment, server computing device **150** and/or home controller **135** may be configured to compare the sensor data received from smart sensors of a home to one or more of the parameters generated using an artificial intelligence model to identify an abnormal level of gas is present in the home, if a gas leak is present, and/or locations of abnormal levels of gas and/or gas leaks within the home. For example, in some embodiments, the parameters may include one or more baseline norms, or expected values or ranges for certain types of sensor data (e.g., gas levels in the air and/or other related types of data), to which server computing device **150** and/or home controller **135** may compare the sensor data to identify an abnormal level of gas is present in the home, if a gas leak is present, and/or locations of abnormal levels of gas and/or gas leaks within the home. For instance, some amount of gas may be normal in certain locations base upon, for example, whether the location is near a gas appliance, whether the area is ventilated, whether the location is in a home, outdoors, or another type of building, etc., but an amount of gas in excess of this normal amount may be a cause for attention and/or taking remedial measures. These baselines norms may be generated by the AI model based upon historical sensor data.

[0057] In some embodiments, certain alert conditions may require a combination of different baseline norms, or that certain length or severity thresholds, or combinations thereof, be met. For example, if a relatively high gas level compared to the baseline norm is detected, but only for a relatively short period of time that does not periodically reoccur, server computing device **150** and/or home controller **135** may record and/or generate alerts for the high gas level but may not necessarily take other measures, such as closing gas shutoff valve **114**. However, if a high level compared to the baseline norm is detected for a threshold period of time and/or if the high level occurs periodically (e.g., whenever a certain gas appliance is first activated), the system may determine there is a higher risk and implement further measures, such as closing gas shutoff valve **114**.

[0058] In some embodiments, the parameters may include one or more digital profiles including a plurality of data values corresponding to different types of sensor data, and the system may compare the sensor data to the plurality of data values of the digital profiles to identify the at least one alert condition. For example, a digital profile may be associated with a presence of abnormal levels of gas, a gas leak, or a certain location of abnormal levels of gas or a gas leak within a home. If the sensor data includes readings that match this digital profile, server computing device **150** and/or home controller **135** may determine that there is a presence of abnormal levels of gas, a gas leak, or a certain location of abnormal levels of gas or a gas leak within home **130**.

[0059] In the exemplary embodiment, server computing device **150** and/or home controller **135** may be configured to control one or more gas shutoff valves **114** within home **130** to close in response to determining that an abnormal level of gas is present in the home. In some embodiments, in which a specific location of abnormal levels of gas and/or a gas leak within the home is determined, server computing device **150** and/or home controller **135** may select certain gas shutoff valves **114** to close based upon the location. For example, if a certain appliance is determined to be leaking, server computing device **150** and/or home controller **135** may determine to close a gas shutoff valve **114** specific to that appliance.

[0060] If abnormal levels of gas and/or leaks are detected in multiple locations, or if gas levels remain present or continue to rise after an initial local shutoff, a main gas shutoff valve **114** to the house may be closed. Examples of fixtures, appliances, or locations that may include gas shutoff valves **114** include a main supply valve (e.g., at the point of entry or gas meter, a gas range, a gas oven, a furnace, a water heater, a fireplace, a gas heater, a gas dryer, an outdoor grill, an external gas portal for lawn or other equipment, or any other gas appliance and/or access point. In some embodiments, certain appliances (e.g., smart appliances and/or IoT devices **110** connected to the home controller) may include integrated gas shutoff valves **114**. Other appliances may have a supply of gas controlled by a gas shutoff valve **114** installed on a gas line leading to the appliance.

[0061] Similarly, in certain embodiments, in response to determining that an abnormal level of gas is present in the home, server computing device **150** and/or home controller **135** may identify one or more ventilation systems (e.g., HVAC, exhaust fans, and/or other fans or controllable vents) capable of dispersing buildups of gas at the location of the gas leak or abnormally high levels of gas and activating these identified ventilation systems.

[0062] In various embodiments, the system may be configured to transmit data (referred to herein as "content data") to a user device **140** associated with home **130** that, when received by user device **140**, causes user device **140** to generate a user interface including any identified alert conditions as to a presence of abnormal levels of gas, a gas leak, or a certain location of abnormal levels of gas or a gas leak within the home and the status of any gas shutoff valves or ventilation systems configured to dissipate gas (e.g., which gas shutoff valves have been closed in response to the detection of abnormal levels, whether ventilation systems were activated, and/or whether such operations were successful). For example, user device **140** may be a mobile device or other computing device capable of executing an app that causes the user interface to be displayed on the user devices **140**.

[0063] In some embodiments, alert notifications may be displayed automatically whenever abnormal levels of gas are detected and/or gas shutoff valves are closed, and nonfictions may also be automatically transmitted to emergency personnel, gas utility companies, and/or any other relevant parties. Additionally or alternatively, the app may include user-selectable options, for example, to contact emergency services or to have gas shut off and an inspection performed by a gas utility company.

[0064] In addition to any identified alert conditions, the user interface may also display other data and recommendations. For example, the user interface to display numbers, indicators, or graphics representing at least some of the received sensor data (e.g., current or historical gas levels), which may illustrate outliers in senor data, locations of detected issues, graphs of norms and current levels, trends over time, comparisons to thresholds considered safe or normal, and comparisons to other homes (e.g., averages in a neighborhood and/or geographic area of the homes).

[0065] In some such embodiments, this data may be continuously or periodically updated to reflect real time conditions within home **130**. The user interface may enable the user to close gas shutoff valves **114** and/or activate ventilation systems remotely from user device **140**.

[0066] In some embodiments, server computing device **150** and/or home controller **135** may be configured to generate at least one recommendation based upon the identified alert condition and cause the user interface to display the generated recommendation. For example, server computing device **150** and/or home controller **135** may identify a certain gas appliance as being a source of a leak and recommend the appliance be replaced, and/or may identify a location of a gas line within the home that is leaking and recommend that emergency and/or repair services may be contacted.

[0067] In certain embodiments, if server computing device **150** and/or home controller **135** determines a risk associated with a particular leak is sufficiently high even if measures such as gas shutoff are implemented, server computing device **150** and/or home controller **135** may automatically contact emergency, gas utility, or other appropriate services. Additionally or alternatively, certain recommendations may be generated prophylactically even if a gas leak is not detected. For example, if a certain model of appliance that is determined to be prone to gas leaks is present in the home, server computing device **150** and/or home controller **135** may generate a recommendation that this appliance be replaced.

[0068] In some embodiments, AI or chatbots may be used to generate recommendations. An output of the AI or chatbot may include computer executable instructions for generating a user interface (e.g., within the app and/or web page) to present the recommendations. The use of the generative AI model may be available in various mediums such as a computer and/or mobile application, chat screens, web page, voice interaction with a voice chat-capable connected home device, voice bots or chat bots, ChatGPT bots, and/or social media messaging. In some embodiments, server computing device **150** and/or home controller **135** may generate reports (e.g., on demand or periodically) indicating any issues that were identified and remedies taken, which may be provided to potential buyers of the home or other parties such as gas utility companies and/or insurers.

[0069] In various embodiments, server computing device **150** and/or home controller **135** (e.g., using an AI model) may search for products, stores, and/or services relating to the recommendations and, in some such embodiments, provide a link (e.g., a hyperlink) to the recommended products, stores, and/or services to the user via a computing device (e.g., via a mobile application, web page, and/or email). For example, the links may relate to products and/or maintenance services that may address alert conditions and/or other issues identified in home **130** (e.g., replacing leaking appliances and/or contact acting repair and/or installation services). The system may use geolocation to determine appropriate products, stores, home services, and/or emergency services located near, at, or around the vicinity of the user's geolocation, which the homeowner may obtain in order to implement recommendations provided by server computing device **150** and/or home controller **135**.

[0070] In some embodiments, the recommendations may include a schedule of actions and/or a curated plan and set of reminders for when certain actions should be performed. Additionally or alternatively, the recommendations may include a list actions to address potential issues identified in home **130**. The recommendations may include a timeline and/or a recommended order for performing these actions. The recommendation list may be presented through a user interface that enables the homeowner to select and/or click on listed actions to automatically purchase and/or schedule products and/or services associated with the actions. For example, if an appliance that leaks gas is identified, the system may provide a link to a repair service and/or to purchase a replacement appliance.

[0071] Server computing device **150** and/or home controller **135** may further provide additional information relating to the recommended actions, such as advantages of the particular actions, costs, potential cost savings (e.g., reducing insurance and/or energy costs), when and how to perform the actions, and/or alternative options. In some embodiments, the recommendations may be generated in response to query by the homeowner and/or may be periodically generated automatically (e.g., as a monthly report to the homeowner).

[0072] In some embodiments, server computing device **150** and/or home controller **135** may be communicably coupled to a communication network and/or a financial services provider. Server computing device **150** and/or home controller **135** may receive insurance information from the financial services provider. Server computing device **150** and/or home controller **135** may connect an insurance policy of the homeowner to the generated recommendations, in which the application may display potential changes to the homeowner's insurance policy based upon, for example, installation of smart sensors **112** in home **130**, a presence of certain conditions, and/or implementation of recommendations that fix and/or otherwise mitigate these conditions. Server computing device **150** and/or home controller **135** may prioritize the recommendations based upon potential changes to a customer's insurance policy or claims information submitted by other customers living in an area geolocated near the homeowner.

[0073] In certain embodiments, server computing device **150** and/or home controller **135** may include a risk evaluation engine that may evaluate data associated with sensor data in a home to evaluate various risks associated with the home. For example, sensor data patterns or profiles may correlate to a certain level of risk of damage occurring to home **130**. Server computing device **150** and/or home controller **135** may use numerous data points to evaluate such risks to a residential property and may compute a composite risk score and/or various focused risk scores for the property. The risk score (e.g., or likelihood of damage score) may be a numeric value and/or a category (e.g., excellent, good, fair, and poor). Such risk scores may be used, for example by an insurance provider, to evaluate insurability of the property and its assets, to price insurance policy options for the property, or to provide policy discounts and verify compliance for risk mitigating changes, actions, or behaviors. Further, such risk scores may be used to determine to recommend actions (e.g., maintenance) be taken.

[0074] In various embodiments, server computing device **150** and/or home controller **135** may generate a risk score for different categories of risk, such as property risk, fire protection, and safety, which may be presented individually within the user interface with related recommendations. For example, the fire protection rating may be displayed along with fire-protection related recommendations, such as recommendation relating to actions that may result in a reduction of fire risk if implemented (e.g., addressing alert conditions identified by the system that relate to fire safety).

Exemplary Home Monitoring System

[0075] FIG. **2** illustrates an expanded computer system **200** that may be used for identifying conditions in home **130** and providing recommendations for improving addressing these identified conditions. In the exemplary embodiment, the system **200** includes server computing device **150** that may be remote from the home. Server computing device **150** may be configured to execute a home monitor and analysis engine **225** and a risk evaluation engine **230**. The server computing device **150** may include or otherwise be in communication with a home analysis database **235** that stores information about the home **130** that may be used in part to identify conditions or issues in home **130**, and may include information about real estate upon which the home **130** is located, assets contained within the home **130**, and various data points that may be used to determine if any alert conditions or issues are present in home **130**. The terms "house," "home," and "residential property" may be used interchangeably herein to refer to the home **130** and its various property and assets.

[0076] In the exemplary embodiment, the server computing device **150** is in networked communication with a home controller (or just "controller") **135** of the home **130** through an external network **210** (e.g., the Internet). The home controller **135** may manage aspects of sensor data collection, computations, and alerting as a part of system **100**. The home controller **135** is connected to a home network **205** of the home **130** which allows communication with server computing device **150** through an external network **210** (e.g., the Internet). For example, the home **130** may include a local area network ("LAN"), a wireless network (e.g., Wi-Fi network), or some combination thereof that connects to the external network **210** (e.g., via a subscription service to an Internet service provider, or the like). In some embodiments, the home controller **135** may communicate via a wireless mobile network, such as a 3G, 4G, or 5G network.

[0077] The home network **205** may allow various devices within the home **130** to communicate over the home network **205**, such as computing devices and Internet-of-Things ("IoT") type devices **110** and/or smart sensors **112** (shown in FIG. **1**). Such IoT devices **110** and/or smart sensors **112** may be referred to herein as "connected home devices," in that they are associated with the home **130** or otherwise a part of the home network **205**. Some IoT devices **110** and/or smart sensors **112** may participate in system **100** and/or system **200**, for example, providing sensor data that may be used (e.g., by server computing device **150**) to evaluate home **130**, to generate risk scores, determine matches in the marketplace server **240**, or other uses described herein.

[0078] In the exemplary embodiment, the systems **100** and **200** may allow homeowners to opt into or out of various aspects of data collection from IoT devices **110** (e.g., by device type, by type of data collected, by data use). For example, the homeowner may be presented with an individual login to the system **100** and **200** which may include an opt-in screen that allows the homeowner to view data collection and usage policy and select whether they wish to allow such usage, thereby protecting privacy of the homeowner. Home data generated by such IoT devices **110** may be referred to herein as just "home data." In some embodiments, the sensor data analyzed by the system may include and/or be a subset of this home data.

[0079] Server computing device **150**, in the exemplary embodiment, may collect some home data from one or more external data sources **215**. The home monitor and analysis engine **225** or the risk evaluation engine **230** may, for example, collect data from publicly available sources or from private third-party sources about the particular subject home **130** or the area in which the home **130** is built (referred to herein as "the locality of the home"). For example, one external data source **215** may be the national weather service ("NWS"), a branch of the national oceanic and atmospheric administration ("NOAA"). The NWS collects, and makes publicly available, weather data for the United States of America and its outlying countries.

[0080] The system **100** and **200** may collect aspects of historical, current, or predictive weather data for a locality of the home **130** (e.g., storm, wind, lightning, flooding in the locality) and may use such data to, for example, evaluate the appliances installed in home **130**. Such data from external data sources **215** is referred to herein as "external data." Some external data sources **215** may maintain such external data in one or more external databases **220**. Other examples of external data sources **215** and external data may be provided by manufacturer server **105** (shown in FIG. **1**) in addition to those provided below, as well as various uses for such external data.

[0081] In the exemplary embodiment, server computing device **150** is in communication with a marketplace server **240** through the external network **210**. The marketplace server **240** is a platform where businesses and/or individuals come together to sell products and services to the customer base of homeowners. The marketplace server **240** and server computing device **150** determine the needs of the users and then determines which product providers **245** (e.g., stores or individuals who have listed products for sale) and service providers **250** that may be of assistance to the user. For example, if an appliance or other device needs to be repaired or upgraded, server computing device **150** may identify a service provider **250** that can perform the needed repair or upgrade and provide a link to communicate with the service provider **250** to the homeowner (e.g., using user device **140**). Similarly, if an appliance or other device needs to be replaced, server computing device **150** may identify a product provider **245** that can provide a replacement and provide a link to communicate with the product provider **245**.

[0082] In the exemplary embodiment, server computing device **150** may be operated by an insurance provider that provides insurance coverage for the home **130** (e.g., via a home insurance policy) or that provides participation in systems **100** and **200** as a home protection service for the homeowner. The insurance provider may be any individual, group of individuals, company, corporation, or other type of entity that may issue insurance policies for customers, such as a homeowners, renters, or personal articles insurance policy associated with the home **130** or an insured. For example, after signing up for a home insurance coverage, the insurance provider may provide the home controller **135** for installation in the home **130**.

[0083] Although the present disclosure describes the systems and methods as being facilitated in part by the insurance provider, it should be appreciated that other non-insurance related entities may implement the systems and methods. For example, a utility company, government or public policy organization, manufacturer and/or general contractor may aggregate the sensor data across many properties to determine which products and/or home configurations provide the best energy efficiency. Accordingly, it may not be necessary for the home **130** to have an associated insurance policy for the property owners to enjoy the benefits of the systems and methods.

[0084] The home controller **135**, as discussed in greater detail below, may be configured to collect home data, such as sensor data, from sensors, appliances, or other devices within the home **130**, connect to the home network **205**, and communicate with server computing device **150** and/or marketplace server **240**. The home controller **135** may be configured to connect to the home network **205** and communicate with other networked IoT devices **110** (or "smart devices") and/or smart sensors **112** within the home **130**. Such IoT devices **110** and/or smart sensors **112** may be referred to herein as "source devices," "connected devices," or "IoT devices," as devices that provide home data to the systems **100** and **200**. In some embodiments, server computing device **150** may communicate directly with some or all of the source IoT devices **110** and/or smart sensors **112** within the home **130**. Various source devices are illustrated in further detail below with respect to FIG. **3**.

[0085] In the exemplary embodiment, server computing device **150** provides the users access to the marketplace, while using ML and AI to determine which product providers **245** and service providers **250** are the most relevant to the user based upon the analysis of home **130**. In at least some embodiments, server computing device **150** determines different attributes and/or conditions of the devices in home **130** based upon the home data provided from IoT devices **110** and/or the external data sources **215**.

Exemplary Source Devices

[0086] FIG. **3** illustrates exemplary source devices that may be used with the system **100** (shown in FIG. **1**) and the system **200** (shown in FIG. **2**). In the exemplary embodiment, the home controller **135** is in communication with or otherwise monitors or collects data from a variety of source devices within the home network **205**. The home **130**, and the various source devices therein, may be powered by an electrical distribution system **300**. Paths of electrical power flow are illustrated in FIG. **3** in broken lines. The electrical distribution system **300** includes multiple electrical circuits **308**, each of which may provide power to one or more of the source devices or other IoT devices **110** within the home **130**.

[0087] Each of the example circuits **308** emanate from an electrical distribution panel **306** that receives power from a power source **310**, such as a utility power company or an on-premise power source (e.g., gas generator, solar generator, wind generator). Each circuit **308** may include a circuit breaker for each circuit **308** in the electrical distribution panel **306**. While not expressly shown, any of the various source IoT devices **110** and/or smart sensors **112** (not shown in FIG. **3**) may be connected to and powered by the electrical circuits **308**.

[0088] In the exemplary embodiment, the systems **100** and **200** may include one or more electricity monitoring ("EM") devices **304**. EM devices **304** may be used to monitor electricity flowing to individual electric devices, such as smart devices or appliances, electronics, vehicles, or mobile devices, and may be configured to monitor or detect abnormal usage or trends. Abnormal electricity flow ("EF") to various devices may indicate that failure is imminent, maintenance or device replacement is needed, de-energization is recommended, or other corrective actions are prudent. For example, the EM devices **304** may be TING® smart sensors such as those made commercially available by Whisker Labs of Germantown, MD.

[0089] EF data collected by the EM devices **304** may include data indicative of electricity flow to or from various smart or other IoT devices **110**, including the various devices shown here in FIG. **3**. EF data may also include electricity or energy usage for each electronic component, device, outlet, circuit, or the like, within the home **130**, such as data indicating the electricity each device or room is using. For example, energy usage of air conditioners, washers, dryers, dish washers, refrigerators, stoves, ovens, microwave ovens, televisions, lamps, outlets, computers, laptops, mobile devices, other electronic devices, may be determined by the EM device **304**.

[0090] In addition to energy usage, EF data may be used to detect hazards or other abnormalities that may be correlated with a reduced energy efficiency of the powered appliances and/or indicate a risk to the home **130** or its assets. For example, changes in electrical consumption (e.g., drawing more power and/or current than usual) of IoT devices **110** may indicate that IoT devices **110** are having problems that may influence an energy efficiency of IoT devices **110**. Accordingly, EF data collected by the EM devices may be fed into the AI model as a factor in determining energy efficiency and generating recommendations to improve the energy efficiency.

[0091] EM devices **304** may include sensors that are configured to monitor and collect EF data. EM devices **304** may be plugged into electrical outlets within the home (e.g., conventional 110-volt outlets) for at least powering the EM device **304** and/or IoT devices **110**, or may be electrically wired into a circuit **308** for powering the EM device **304** and/or IoT devices **110**. Further, some EM devices **304** may collect EF data directly from a circuit **308** (e.g., via wired connection to the circuit **308**, referred to herein as "direct sensing") and some EM devices **304** may wirelessly collect EF data from circuits **308**, appliances, or other electricity consuming devices (referred to herein as "wireless sensing"). Wireless sensing may include, for example, sensors within the EM device **304** that are configured to sense electromagnetic waves or an electrical signature of the electrical devices receiving power from the electrical distribution system **300**. The EM devices **304** may directly or wirelessly detect each flow of electricity to or from each different electronic device by identifying each electronic device by its unique electronic or electrical signature (or "fingerprint").

[0092] The EM devices **304** may then generate electricity usage or flow data for each electronic device within the home, or connected to the electrical distribution system **300** (such as a hybrid or fully electric vehicle having its battery directly or wirelessly charged by the home's electrical system). In some embodiments, EM devices **304** may be positioned in vicinity of the electrical distribution panel **306** and may capture electrical activity about the home **130** and/or devices installed in the home **130** by wirelessly detecting an electricity flow to devices that are coupled to the electrical distribution panel **306**.

[0093] In other embodiments, EM devices **304** may be positioned in vicinity of the electrical distribution panel **306**, but not hardwired to the electrical distribution panel **306** or home electrical wiring system, and may capture electrical activity about the home **130** and/or appliances installed in the home **130** by wirelessly detecting an electricity flow to devices that are coupled to the electrical distribution panel **306**. In other embodiments, EM devices **304** may be plugged into electrical outlets positioned throughout a home.

[0094] During operation, as one or more of the electric devices receives electricity via the electrical distribution system **300**, each device may be differentiated by an electrical signature that is unique to a respective device (such as by one or more EM devices **304** monitoring, detecting, and/or analyzing the electricity flowing to or being consumed by each respective electric device, and/or by monitoring EF data generated or collected by one or more EM devices **304**).

[0095] In other words, transmission of electricity to a refrigerator, for example, may be differentiated from transmission of electricity to an electric stove (such as via one or more EM devices **304** and/or analyzing the EF data generated or collected by one or more EM devices **304**). Furthermore, transmission of electricity to a television on one circuit **308** or outlet, for example, may be differentiated from transmission of electricity to another recipient electric device (e.g., a cable television box) via the same circuit **308** or electrical outlet. The systems **100** and **200** may correlate electrical activity with a variety of electric devices on the electrical distribution system **300** based upon electrical signatures unique to each respective device.

[0096] The systems **100** and **200** may build a structural electrical profile for the home **130**, which may include data indicative of operation of the various electric devices within or around the home **130** (e.g., over a period of time), such as by using EF data generated or collected by one or more EM devices **304** over a period of time. In some embodiments, the electrical profile may further be used in identifying specific models of appliances to be added to the digital home profile.

[0097] In certain embodiments, an EM device **304** may be affixed to or situated near the electrical distribution panel **306**. Generally, the EM device **304** may utilize the unique, differentiable electrical signatures of the electric devices by directly or wirelessly monitoring electrical activity including transmission of electricity via the electrical distribution panel **306** to one or more of the electric devices. Monitoring of transmission of electricity to an electric device receiving the electricity may include, for example, monitoring (i) the time at which the electricity was transmitted, (ii) the duration for which the electricity was transmitted, and/or (iii) the magnitude of the electric current in the transmission.

[0098] Based upon the unique electrical signatures of the various electric devices of the home **130**, the monitored electrical activity may be correlated with respective electric devices receiving the electricity transmitted via the electrical distribution system **300**, enabling the electricity usage of the various devices to be tracked individually. Further, electrical activity associated with other components of the electrical distribution system **300** (e.g., the electrical distribution panel **306**, the circuits **308**, or the like) may be correlated with one or more electric devices to which the electrical activity also pertains.

[0099] In some embodiments, the EM device(s) **304** may perform the correlation or other functions described herein, via one or more processors of the EM device(s) **304** that may execute instructions stored at one or more computer memories of the EM devices **304**. In other embodiments, the EM devices **304** may collect the EF data, and the correlation and/or other functions described herein may be performed at another system (e.g., the home controller **135** or server computing device **150**), which may receive data or signals indicative of monitored electricity or other data via one or more processors or through transfer via a physical medium (e.g., a USB drive). Correlation of the electrical activity with the respective electrical devices may produce data indicating, for example, the time, duration, and/or magnitude of electricity consumption by each of the electric devices during a period of electrical activity monitoring.

[0100] Based upon at least the correlated electrical activity, a structure electrical profile may be built and stored at the EM devices **304** or at some other system (e.g., the home controller **135** or the home analysis database **235**). The structure electrical profile may include, for each of the electric devices about the home **130**, data indicative of operation of the respective electric device during at least the period at which the EM devices **304** monitored electrical activity about the home **130**. Based upon the correlated electrical activity, the structure electrical profile may depict, for example, average electricity operation/usage, baseline electricity operation/usage, and/or expected electricity operation/usage/consumption. In effect, the structure electrical profile, based upon electrical activity about the structure, may set forth what is "normal" operation and usage of electricity about the structure.

[0101] Thus, once the structure electrical profile is built, any electrical activity monitored via the home controller **135** and the EM device(s) **304** may be analyzed to determine whether electrical activity is abnormal and/or otherwise indicative of a condition that my affect the energy efficiency of the electric devices. In response to the abnormal electrical activity, among other possible factors, corrective actions to improve the energy efficiency of the device, mitigate damage, prevent damage, and/or remedy the cause of the abnormal electrical activity the situation may be determined and/or initiated. Some possible corrective actions are discussed herein.

[0102] EF data regarding an electric device may include, for example, historical data indicating the electric device's past operation patterns or trends. For example, historical data may indicate a time of day, day of the week, time of the month, etc., at which an electric device frequently uses electricity (e.g., a lighting fixture may not use electricity during late night hours of the day). As another example, historical data may include the electric device's total electricity consumption or usage rate over a period of time. Additionally or alternatively, historical data may include data indicating past events regarding the electric device (e.g., breakdowns, power losses, arc faults, etc.).

[0103] Additionally or alternatively, operation data regarding an electric device may include an expected electricity consumption or baseline electricity consumption for the electric device. For example, in the case of a refrigerator, the refrigerator's electricity consumption during a first period of monitoring may be reliably used to approximate an expected electricity consumption at a later time. Changing electricity consumption over time (e.g., the refrigerator's consumption is greater than expected for a period) may indicate that the refrigerator is in need of repair and/or maintenance and/or operating sub-optimally.

[0104] Further, the structure electrical profile may include data pertaining to the structure as a whole. For example, the structure electrical profile may include data reflecting a total electricity or average usage rate over a period of time. As another example, the profile may include time-of-day, day-of-week, etc., data reflecting times at which the home **130** as a whole uses more or less electricity. Further, the profile may detail specific types, classes, or specifications of electric devices that behave differently or consume a different amount of electricity compared to other electric devices within the home **130**. Further, the profile may detail specific risks determined to be relevant to one or more of the electric devices or to the home **130** as a whole, based upon the electrical activity of the electric devices.

[0105] Furthermore, the structure electrical profile may include a digital "map" of the home **130**. A home map may indicate spatial locations of the electric devices, and/or spatial relationships between two or more of the electric devices. Such mapping may indicate, for example, energy efficiency implications and/or a risk associated with the spatial placement of a stove, and/or energy efficiency implications and/or a risk associated with placing a refrigerator adjacent to the stove.

[0106] Additionally or alternatively, the home map may indicate which of the electric devices are connected to each electrical circuit **308** within the electrical distribution system **300** of the home **130**. Such mapping may indicate, for example, a risk of overloading a particular circuit **308** based upon a number or intensity of electric devices connected to the circuit **308**. As another example, the home map may be used to determine what electric devices may lose power if a particular circuit **308** were to be de-energized (e.g., due to risk or abnormal electrical activity associated with one electric device on the circuit).

[0107] In some embodiments, the home map may be configurable by a user (e.g., the homeowner of the home **130**). The user may, for example, configure the map via an I/O module (e.g., screen, keypad, mouse, voice control, etc.) of the home controller **135**, or via an I/O module of another computing device, which may transmit the home map to the home controller **135**. Additionally or alternatively, the home map may be stored at one or more computer memories of another system (e.g., server computing device **150**).

[0108] In some embodiments, the home network **205** may include a home power management system **326**. The home power management system **326**, or home controller **135** in conjunction with the EM devices **304**, may collect power consumption data on the circuits **308** (e.g., via EM devices **304**) or device electrical usage data of various electronic devices within the home **130**. The home power management system **326** may, for example, collect usage data for lights or appliances within the home **130**, giving an indication of how much electricity the home **130** uses or how frequently occupants are at home. In some embodiments, the home **130** may include one or more smart plugs (not separately shown) which may be managed by home power management system **326**, the smart speaker device **318**, the smart home system **324**, or otherwise by the systems **100** and **200** (e.g., for activating or deactivating devices plugged into the circuits **308** via the smart plugs, such as via 110-volt outlets).

[0109] The home power management system **326** may identify and provide details on what appliances or other consuming devices are within the home **130** (e.g., manufacturer make and model), thereby allowing the systems **100** and **200** to identify some property on the premises (e.g., device identification and verification, device count), evaluate value of devices (e.g., replacement costs), or collect manufacturer-provided or consumer protection-provided details regarding the devices from external data sources **215** (e.g., susceptibility of the device to power surges, likelihood of fire caused by the device, mean time to failure of the device, types of device failures, power consumption profiles and tolerances of the device, or the like).

[0110] The home power management system **326** may collect power quality data for the home **130**, such as occurrences and frequency of power outages or reductions in service (e.g., black-outs or brown-outs), loading at various times throughout the day or week, the size of service, occurrences of voltage values fluctuating beyond tolerance ranges (e.g., spikes), or the like. In some embodiments, the home power management system **326** may include one or more smart circuit breakers (e.g., on any or all of the circuits **308**) or a smart panel (e.g., as the electrical distribution panel **306**), such as those made commercially available by Schneider Electric (Paris, France), which may provide circuit-level data and operations such as, for example, current or historical circuit load data, circuit breaker status, or turning circuit breakers on or off. Such power data may be used to construct a power profile for the home **130**. In some embodiments, the home controller **135** may perform any such power monitoring and data collection operations in lieu of, or in addition to, the home power management system **326**.

[0111] In the exemplary embodiment, the home **130** may include one or more smart appliances **312** (e.g., appliances that can communicate via the home network **205**, which may include IoT devices **110**). Smart appliances **312** may include, for example, dish washers, microwaves, stove tops, ovens, grills, clothes washers and dryers, water heater, water meter, water softener or purifier, smart lighting, smart window blinds or shutters, piping, interior or yard sprinklers, or the like. The home controller **135** may be configured to communicate with such smart appliances **312** and may collect home data from such appliances for the systems **100** and **200**.

[0112] For example, smart appliances **312** may provide data such as device data (e.g., manufacturer, make, model, date of manufacturer, date of installation, software or firmware versions), usage data (e.g., daily usage time, power consumption), or log data (e.g., log events, alerts, component failure detections, maintenance history, or the like). Such appliance data may allow the systems **100** and **200** to detect which appliances are present in the home **130** (broadly, as a part of an "asset inventory" of the house), their replacement value, age of each appliance, a maintenance history of each appliance, to detect when appliances or their components are failing.

[0113] Electrical distribution system **300** may use such data, for example, to construct the power profile for home **130**, to compute a risk for the home **130** and/or the appliances, to compute in an insurance profile for the home **130** (e.g., as factors of risk to lightning or other hazards), or to alert the homeowners when an appliance registers a failure.

[0114] In the exemplary embodiment, the home **130** may also include smart HVAC devices such as, for example, a heater (e.g., a gas or electric furnace), an air conditioner, an air purifier, an attic fan, a ceiling fan. Some or all such devices may be controlled by a thermostat device. Such devices are collectively referred to herein as HVAC devices **314**, some of which may not be smart devices but may nonetheless be controlled in some aspects by the thermostat device.

[0115] The systems **100** and **200** may collect HVAC data such as device data (e.g., manufacturer, make, model, date of manufacturer, date of installation), usage data (e.g., daily usage time, power consumption), or thermostat data (e.g., temperature settings, daily schedule profiles). The systems

100 and 200 may use such data, for example, to construct the power profile for the home 130, to compute a risk for the home 130 (e.g., determining how often the home 130 is typically occupied), to compute in an insurance profile for the home (e.g., as factors of risk to lightning or other hazards, likelihood of equipment failures), or to alert the homeowners when an HVAC device registers a failure.

[0116] The home **130**, in the exemplary embodiment, may also include various computing devices such as, for example, desktop or laptop personal computers, tablet computers, servers, or networking devices (e.g., Wi-Fi routers, switches, hubs, firewalls, or the like), all of which are collectively represented here as home network/computer devices (or just "computer devices") **316**. The networking devices may provide some or all of the home network **205** that is used to facilitate communication between the devices shown here. The home controller **135** may be configured to capture computer device data from some or all of these home network computer devices **316** such as, for example, a number and type of computing devices (e.g., hardware manufacturer, make, model, and the like), hardware and software profile of computing devices, configuration data of computing devices (e.g., software versions, firmware versions), usage data, and log data (e.g., firewall logs, access logs, software patch logs, error logs). The systems **100** and **200** may use such data to, for example, determine asset inventory and valuation, construct the power profile for the home **130** (e.g., average daily usage), alert the homeowners when devices need software or firmware upgrades (e.g., critical security alerts) or upon intrusion detection or other compromise of home network computer devices **316** (e.g., software hacks).

[0117] In the exemplary embodiment, the home **130** may include a smart speaker device(s) (or "nest device") **318** that may interact with occupants of the home **130** (e.g., via audible commands and responses, digital display, executing pre-configured actions). Some example smart speaker devices **318** include the Echo® devices (Amazon Inc., of Seattle, Washington) and the Google Nest® devices (Alphabet Inc., of Mountain View, California), to name but a few. The smart speaker device **318** may include a speaker for providing audio output, a microphone for receiving audio input (e.g., commands spoken by the occupants), and may include a display device for video output or a camera device for capturing video input. The smart speaker device **318** may be configured to interact with other smart devices, such as for controlling lighting within the home **130**, the thermostat (e.g., changing thermostat settings), home security devices of a home security system **320** (e.g., locking and unlocking smart locks on doors, opening or closing garage doors, or the like), or entertainment devices of a home entertainment system **326** (e.g., enabling, disabling, or reconfiguring music or television devices).

[0118] The systems **100** and **200** may, with owner configuration and permission, utilize inputs from the smart speaker device **318** to, for example, determine a number of unique occupants of the home **130** (e.g., via unique speech profile or video identification), determine the number of children in the home **130** (e.g., via audio or video analysis), determine when occupants of the home **130** are currently or historically present (e.g., via noise detection, video movement), determine when other devices are turned on or off, determine presence of pets (e.g., via unique audio sounds or video identification of the pets), or smoke or carbon monoxide alarm detection (e.g., via audible sound). Such raw data may be sanitized or distilled by the home controller **135** into refined data before sending to server computing device **150** in an effort to protect privacy of the home occupants while still providing home health evaluation and risk capabilities (e.g., sending results determined from the raw audio or video data and deleting the raw audio or video data). The systems **100** and **200** may anonymize personal data, thereby allowing data to be stored or used without direct attribution of data to a particular homeowner.

[0119] In the exemplary embodiment, the home **130** may include various home entertainment devices **320** such as, for example, televisions, digital video recorders ("DVR"), radios, amplifiers, speakers, remotes, or console gaming systems, any or all of which may be smart devices in communication with the home network **205** and home controller **135**. Home controller **135** may collect home entertainment data from such devices and may use that data, for example, to construct the power profile for the home **130**, to construct the asset inventory of the home **130**, to compute a risk score for the home **130**, to compute in an insurance profile for the home (e.g., as factors of risk to lightning or other hazards, likelihood of equipment failures).

[0120] The home **130**, in the exemplary embodiment, may include a home security system **322**. The home security system **322** may include security devices such as, for example, door or window sensors (e.g., to detect when doors or windows or open, when windows are broken), motion sensors (e.g., to detect when someone is present within range of the sensor), security cameras (e.g., for capturing audio/video of particular areas in or around the home **130**, such as a doorbell camera), key pads (e.g., for enabling/disabling the security system), panic buttons (e.g., for alerting a security service or authorities of an emergency situation), security hubs (e.g., for integrating individual security devices into a security system, for centrally controlling such devices, for interacting with third parties), electric door locks, or smoke/fire/carbon monoxide detectors. Such "security devices" broadly represent devices that can detect potential contemporaneous risks to the home **130** or its occupants (e.g., intrusion, fire, health).

[0121] The home security system **322** may be configured to communicate with a third-party security service or local authorities, and may transmit alerts to such parties when events are detected. The home controller **135** may be configured to receive alert data from the home security system **322** and may transmit such alerts to server computing device **150**, create historical logs of security events, or transmit alert events directly to the homeowner (e.g., via SMS text message or the like) or to local authorities, fire protection, or emergency services. The systems **100** and **200** may use such security alert events to, for example, determine how frequently security events occur (e.g., as a factor for risk), how often such events are warranted (e.g., authentic risks rather than false alarms), or the type and nature of such authentic risks or false alarms.

[0122] The systems **100** and **200** may use raw data collected directly from any of these security devices. For example, the home controller **135** may use raw data from the motion sensors to detect when the home **130** is occupied (e.g., to build a profile of occupancy times), may use raw data from the camera devices or door devices to detect when occupants enter or exit the home **130**, may use the camera devices to determine a number of occupants of the home **130** or a number and type of pets in the home **130**.

[0123] The home controller **135** may determine information about the home security system **322** installed within the home, such as a number and type of security sensors installed within the home **130**, a type of home security system **322** installed in the home (e.g., third-party service provider, device manufacturers, types of security protection implemented within the home), or how often the homeowners leave the home **130** unoccupied without activating the home security system **322** (e.g., as a factor in risk calculations or home health scoring). The systems **100** and **200** may rate the home security system **322** and associated devices and services to generate a home security protection rating (e.g., relative to other available security systems or hardware) and may use that rating as a factor in risk calculations or in preparing a risk mitigation proposal (e.g., for more or better devices or security systems).

[0124] In some embodiments, the home **130** may include a smart home system **324** (e.g., a home monitoring system) that allows the homeowner and occupants to control various devices within the home **130**. For example, the smart home system **324** may be configured to control, inter alia, devices such as the smart appliances **312**, HVAC devices **314**, home entertainment devices **320**, or home security system **322**. In the exemplary embodiment, the home controller **135** may be configured to interact directly with such devices as described herein ("direct access"), or may be configured to perform some interactions and data collections with such devices through the smart home system **324** ("proxy access"). For example, any or all of the data collections or operations described herein may be performed by the smart home system **324** based upon commands received from the home controller **135**, thereby allowing the systems **100** and **200** to perform such operations through the smart home system **324** acting as a proxy for some such operations.

[0125] In the exemplary embodiment, the home **130** may include a home car charging station **328** that may be used to recharge electric vehicles. The home car charging station **328** may draw power from one or more of the circuits **308** of the electrical distribution system **300** and may include an on-premise power source (e.g., solar panels, wind generator, or the like) or a dedicated battery bank (e.g., for storing excess power from the local energy source). The systems **100** and **200** may capture various charging station data from the home car charging station **328**, from the circuits **308** used for home car charging station **328**, or from the local power source device(s).

[0126] In the exemplary embodiment, the home **130** may include one or more smart alarms **330** that are configured to detect various conditions within the home **130** and may alert the homeowner or other occupants (e.g., via audible alarm, SMS text message, email, or the like). Smart alarms **330** may include, for example, smoke detectors, carbon monoxide detectors, carbon dioxide detectors, or indoor air quality ("IAQ") monitors or systems that include sensors configured to, for example, detect dangerous conditions such as fire or buildup of carbon monoxide, the presence of dangerous pollutants such as radon or various volatile organic compounds ("VOC"), or collect various air quality data such as temperature and humidity. Smart alarms **330** may include water leak detectors or flood alarms that may be configured to detect the presence of water at various areas in the home **130**, such as near HVAC equipment, water tanks, sump pumps, below showers or bathtubs, around basement perimeters, behind or within basement walls, or the like. Such water detectors may identify leaks within plumbing or appliances within the home **130** or ingress of water into the home **130** (e.g., rain water, flooding, failing sump pump, foundation cracks, or the like).

[0127] System **100** may collect alarm data from the smart alarms **330** and may perform automatic alerting based upon sensor events registered at such smart alarms **330** (e.g., alerting emergency services, homeowner, or the like, in an effort to protect life and property, mitigate damage, or such) or initiate automatic actions (e.g., shutting off water flow within the home **130**, or within a particular segment of plumbing, via activating a smart water shut off valve, not separately shown). The systems **100** and **200** may identify the presence of such smart alarms **330** or shut off valves in the home **130** when configured to communicate with the smart alarms **330** and may automatically provide policy discounts when particular smart alarms **330** are detected as present or may include the presence or absence of such smart alarms **330** in the various aspects of home health scoring. Furthermore, server computing device **150** may be configured to provide marketplace suggestions of provides to assist with the issues that are associated with the alarms.

[0128] Data received from smart alarm **330** may be used to detect hazards or other abnormalities that may be correlated with a reduced energy efficiency of the powered appliances, a need to repair or replace certain IoT devices **110**, and/or indicate a risk to home **130** or its assets. For example, if smart alarm **330** is triggered based upon poor air quality in home **130**, it may be determined that there is an issue with certain appliances such as HVAC devices **314**, fans, and/or air purifiers, which may be correlated with suboptimal energy efficiency.

Exemplary External Data Sources

[0129] In the exemplary embodiment, and referring now to FIG. **2**, the system **200** may collect various types of external data from external data sources **215** that may be used, for example, for evaluating home **130**, or other various uses described herein. For example, the machine learning model or AI model may identify correlations between any of the data types described herein and alert conditions in a home, and therefore may use any of these data sources as factors in identifying alert conditions in a home. Some external data sources **215** may provide publicly available data, where other external data sources **215** may be private, third-party sources. External data sources **215** may include an insurance provider that provides insurance policies to the homeowner and various data available or otherwise collected by that insurance provider. In some embodiments, server computing device **150** may be operated by the insurance provider and the data may include data private to the insurance provider (e.g., customer data, policy information, or other proprietary information).

[0130] In the exemplary embodiment, one example external data source **215** is the NOAA or any of its various branches (e.g., the national weather service). The NOAA makes various weather data publicly available. As such, the system **200** may collect weather data from the NOAA. Such weather data may be refined to a particular geography, such as a state, county, city, or other geographic region. The system **200** may, for example, identify a geographic region of the home **130** and submit data queries to the NOAA for weather data specific to that geographic region. Such data queries may include requests for historical data such as average rainfall, storm occurrences, wind strengths, lightning strikes, temperatures, tornado events, or the like. Data queries may include requests for forecast data such as severe watches warnings, tornado watches or warnings, flooding watches or warnings, precipitation predictions, wind predictions, lightning event predictions, blizzard warnings, or the like. Forecast data may be used to, for example, generate and send weather alerts to the homeowner or occupants of the home **130** or determine how frequently the home **130** experiences various warnings or alerts over time. In some embodiments, the machine learning model or AI model may identify correlations between weather data and certain conditions, such as a presence of abnormal gas levels and/or gas leaks, that may be present in home **130**, and therefore may use such data as a factor in identifying such conditions.

[0131] In the exemplary embodiment, another example external data source **215** may be the U.S. Forest Service. The U.S. Forest Service maintains historical data related to forest fires and tracks active forest fires in the United States. As such, system **100** may collect forest fire data from the U.S. Forest Service. Such forest fire data may similarly be refined to a particular geography, such as a state, county, city, or other geographic region. The system **200** may, for example, collect historical forest fire data for the geographic region of the home **130**, or may collect current forest fire data at or near the location of the home **130** (e.g., within a pre-defined distance from the home, within a distance from a projected path of the forest fire). System **200** may use current forest fire data to, for example, generate and send forest fire alerts to the homeowner or occupants of the home **130**, or as factors in home health scoring. In some embodiments, the machine learning model or AI model may identify correlations between forest fire data and certain conditions, such as a presence of abnormal gas levels and/or gas leaks, that may be present in home **130**, and therefore may use such data as a factor in identifying such conditions.

[0132] In the exemplary embodiment, another example external data source **215** may be municipal power utilities. Electrical distribution system **300** may access current or historical power network data provided by power utility companies in various localities, such as power generation performance statistics (e.g., generation and load statistics), power transmission and distribution statistics or power outage information (e.g., across the network, local to a distribution segment that services the home **130**, consistencies of voltages, power sags, power surges, brown-outs or blackouts and associated frequencies or lengths of outages, or the like), lightning strike data affecting the power network, or electrical consumption data for the home **130** (e.g., current or historical power usage, local power generation provided back to the network). System **100** may use current power network data to, for example, generate and send alerts to the homeowner during power outages (e.g., as SMS text messages or emails that can be viewed on mobile computing devices), or as factors in identifying alert conditions in a home and/or generating recommendations. In some embodiments, the machine learning model or AI model may identify correlations between power network data and certain conditions, such as a presence of abnormal gas levels and/or gas leaks, that may be present in home **130**, and therefore may use such data as a factor in identifying such conditions.

[0133] In the exemplary embodiment, another example external data source **215** may be third-party home data systems such as Multiple Listings Service ("MLS"), Zillow (www.zillow.com), or other Internet-accessible sources for property data. The system **200** may access such home data systems to collect construction details about the home **130** such as, for example, the age of the home, how many bedrooms and bathrooms the home **130** has, the type of any HVAC, the square footage of the home **130**, the size of the property, market price of the home, whether the home **130** is constructed of wood, brick, concrete, or the like, the type and size of any garage, the quality of materials used to construct the home **130**, whether the home **130** has a basement, the type, age, or condition of plumbing or wiring inside and outside the home **130**, whether the home **130** has a pool and safety fence around the pool, the type of roofing, the floor plan, the architecture of the home **130** (e.g., ranch, two story, split foyer), the type of flooring, the type of exterior (e.g., wood, brick, siding), type of local power generation on the property (e.g., solar, wind, generator), number of fire places, type of fencing or gutters, whether the home **130** has a pool, sheds, patios, porches, or other exterior structures, whether the home **130** has outside doors having steps, type of ducting and insulation within the home **130**, type of landscaping around the home **130**, or mobility or accessibility options within the home **130**.

[0134] Some home statistics data may include geographic data about the home **130** such as, for example, school district information (e.g., public school system, school ratings), utility providers available to at the location (e.g., electric, gas, sewer, waste, recycling, phone, Internet, television, fire, police, hospital, or other city services), proximity data to various services and amenities (e.g., distances from schools, parks, grocery, gas, library, or sources of entertainment), hazard data for the area (e.g., crime statistics, natural disaster statistics, ratings for emergency services). Some home statistics data may include historical data, such as price history (e.g., sales history, listings history), public tax history, insurance claims history, home warranty information, home inspection information, lease information (e.g., whether and how often the home **130** has been partially or fully rented or leased), or the like. Some home statistics data may include home data such as, for example, whether the home **130** is energy certified, type and size of power generation, home appliance or lighting energy certification data, or the like. In some embodiments, the machine learning model or AI model may identify correlations between property data and/or home statistics data and certain conditions, such as a presence of abnormal gas levels and/or gas leaks, that may be present in home **130**, and therefore may use such data as a factor in identifying such conditions.

[0135] In the exemplary embodiment, another example external data source **215** may be an insurance provider or other service provider that has an economic or consumer relationship with the homeowner. The system **200** may access the service provider systems to collect demographic details about the home **130** and its occupants, such as, for example, names or ages of the occupants, education levels or occupations of the occupants, whether any of the occupants smoke, a family emergency plan, community engagement of the occupants, or whether a business is operated out of the home **130**.

[0136] The service provider system may collect home maintenance data about the home **130** such as, for example, maintenance logs of operations performed on the home **130** (e.g., service calls, property damage and fixes, routine device maintenance, cleanings, bug or pest service, lawn or garden service, roofing replacement, or the like), equipment installations and removals, device warranty information, or home improvements (e.g., new deck, pool, room(s), interior or exterior painting or weather proofing, solar installation, water reclamation systems installation, room remodeling, or the like).

[0137] The service provider system may collect home configuration data about the home **130** such as, for example, whether GFCI outlets or LED lights are installed in the home **130**, whether power strips supporting multiple devices are in use, whether the home **130** has exercise equipment, types of grills or fryers installed in the home **130**, whether the home **130** includes particular safety equipment (e.g., smoke or carbon monoxide detectors, fire extinguishers, deadbolts on exterior doors, water sensors, sump pump, or the like), paint colors used on various walls of the home **130**. In some embodiments, the machine learning model or AI model may identify correlations between maintenance data and certain conditions, such as a presence of abnormal gas levels and/or gas leaks, that may be present in home **130**, and therefore may use such data as a factor in identifying such conditions.

[0138] In some embodiments, the service provider may be the operator of server computing device **150** and the homeowner may provide such data via an input interface (e.g., online questionnaire, user interface, service application, or the like, during participation in the home health system described herein). Collection and use of such data may be opted into by the homeowner on behalf of the occupants. Additionally or alternatively, the system **200** may query the homeowner for any data elements described herein and not otherwise automatically accessed by the system **200**.

[0139] In the exemplary embodiment, the system **200** may access aerial data of the home **130**, such as satellite-, aerial-, or drone-captured overhead images of the home **130** and surrounding property. Such aerial data may be used to determine various externally visible features of home data (e.g., via digital image processing, machine learning, or human analysis). For example, system **200** may use aerial data to determine structural elements of the home **130** or surrounding property, such as whether the home **130** has a swimming pool, a fence, or a deck, how many garages the home **130** has, or the like. The system **200** may use aerial data to determine whether the home **130** has trees nearby (e.g., which may cause damage to the home **130**) or whether the home **130** is located on a cul-de-sac or a busy road.

[0140] Such aerial data may be provided by a third party or public external data source **215** (e.g., United States Geological Survey ("USGS"), National Aeronautics and Space Administration ("NASA"), NOAA, Google®, or the like) or may be privately collected (e.g., via aerial or drone photography of the home **130** by the insurance provider, realtor, or the like). Such aerial data may include global positioning system ("GPS") location data for the home **130**. In some embodiments, the machine learning model or AI model may identify correlations between aerial data and certain conditions, such as a presence of abnormal gas levels and/or gas leaks, that may be present in home **130**, and therefore may use such data as a factor in identifying such conditions.

[0141] The system **200** may train a model of satellite images of homes **130** with labeled data of the homes **130** indicating, for example, whether the homes **130** have pools, decks, nearby trees, or other such features. As such, the trained model may be configured to automatically evaluate an unlabeled home (e.g., the home **130** in FIG. **1**) to determine whether such features are present or otherwise categorize the home **130** with respect to those features.

[0142] In certain embodiments, the system **200** may access mapping data around the home **130** to determine various home health features. The system **200** may utilize a web mapping service (e.g., Google® Maps or the like) as an external data source **215**. For example, the system **200** may access the web mapping service via an application programming interface ("API") that allows system **200** to submit, for example, the postal address of the home **130** or a GPS coordinate of the home **130** and query the web mapping service to provide features such as distances to nearby services (e.g., distance to nearest hospital, fire department, police station, schools, places of worship, parks, grocery stores, to various types of entertainment or other amenities, or the like). Mapping data may be used to determine whether the home **130** is situated on a busy or isolated road. The system **200** may generate a play score for the home **130** using the mapping data, where the play score evaluates proximity of the home **130** to various types of entertainment or exercise venues, such as proximity to hiking trails, bike paths, sports fields, professional sports venues, restaurants, theaters, or the like).

[0143] The mapping data may include ground-level imagery provided by the web mapping service that may be used by the system **200** to evaluate various externally visible features of home data (e.g., via digital image processing, machine learning, or human analysis). For example, the system **200** may use ground-level imagery to determine structural features of the home **130** such as a number of stories of the home, type of windows installed in the home, a roof type or type of exterior of the home, or how many garages the home has.

[0144] The system **200** may train a model of ground-level images of homes **130** with labeled data of the homes **130** indicating, for example, how many stories or garages the homes **130** have, what type of exterior or roof type the homes **130** have, or other such features. As such, the trained model may be configured to automatically evaluate an unlabeled home (e.g., the home **130** in FIG. **1**) to determine whether such features are present or otherwise categorize the home **130** with respect to those features. In some embodiments, the machine learning model or AI model may identify correlations between mapping data and certain conditions, such as a presence of abnormal gas levels and/or gas leaks, that may be present in home **130**, and therefore may use such data as a factor in identifying such conditions.

Exemplary Server Computing Device

[0145] FIG. **4** is a schematic diagram illustrating further detail of server computing device **150** (shown in FIG. **1**). Server computing device **150** may communicate with other components of system **100**, such as manufacturer server **105**, IoT devices **110**, smart sensors **112**, home controllers **135**, and/or user devices **140**, via a network **400**. Server computing device may include and/or be in communication with a database **402** that stores data **404** including historical data and other information relevant to identifying alert conditions in home **130** and/or generating recommendations. Data **404** received from network **400** may be stored in database **402**.

Server computing device **150** may configured to use data **404** to generate an operational predictive model module **406** for generating comparison parameters and/or generating recommendations, as described above.

[0146] In exemplary embodiments, server computing device **150** includes a training set builder module **408** configured to submit one or more queries **410** to database **402** to retrieve subsets **412** of data **404**, and to use those subsets **412** to build training data sets **414** for generating operational predictive model **206**. For example, query **410** may be configured to retrieve certain fields from data **404** for homes (e.g., home **130**) having certain similar aspects, such as similar sizes, ages, building materials, and/or locations.

[0147] In exemplary embodiments, training set builder module **408** may be configured to derive training data sets **414** from retrieved subsets **412**. Each training data set **414** corresponds to a historical data **404** ("historical" in this context means completed in the past, as opposed to completed in real-time with respect to the time of retrieval by training set builder module **122**). Each training data set **414** may include "model input" data fields along with at least one "result" data field representing historical feedback, such as reports relating to conditions of homes **130**, feedback received from homeowners, and/or decisions made by homeowners based upon previous recommendations (e.g., whether homeowners performed recommended actions). The model input data fields represent factors that may be expected to, or unexpectedly be found during model training to, have some correlation certain conditions that may occur at home **130**.

[0148] In certain embodiments, the model input data fields in training data sets **414** may be generated from data fields in subset **412** corresponding to historical data **404**. In other words, a trained machine learning model **416** produced by a model trainer module **418** for use by operational predictive model module **406** is trained to make predictions based upon input values that can be generated from the data fields in data **404**. Values in the model input data fields may include values copied directly from values in a corresponding data field in the retrieved subset **412**, and/or values generated by modifying, combining, or otherwise operating upon values in one or more data fields in the retrieved subset **412**. Values in the model input data fields may include sensor data and/or other types of home data. The use of such data fields as model input data fields facilitates the machine learning model in weighing these factors directly.

[0149] After training set builder module **408** generates training data sets **414**, training set builder module **408** passes the training data sets **414** to model trainer module **418**. In example embodiments, model trainer module **418** is configured to apply the model input data fields of each training data set **414** as inputs to one or more machine learning models. Each of the one or more machine learning models is programmed to produce, for each training data set **414**, at least one output intended to correspond to, or "predict," a value of the at least one result data field of the training data set **414**. "Machine learning" refers broadly to various algorithms that may be used to train the model to identify and recognize patterns in existing data in order to facilitate making predictions for subsequent new input data.

[0150] Model trainer module **418** may be configured to compare, for each training data set **414**, the at least one output of the model to the at least one result data field of the training data set **414**, and apply a machine learning algorithm to adjust parameters of the model in order to reduce the difference or "error" between the at least one output and the corresponding at least one result data field. In this way, model trainer module **418** trains the machine learning model to accurately predict the value of the at least one result data field. In other words, model trainer module **418** cycles the one or more machine learning models through the training data sets **414**, causing adjustments in the model parameters, until the error between the at least one output and the at least one result data field falls below a suitable threshold, and then uploads at least one trained machine learning model **416** to operational predictive model module **406** for application to generating predictions **420**. In exemplary embodiments, model trainer module **418** may be configured to simultaneously train multiple candidate machine learning models and to select the best performing candidate for each result data field, as measured by the "error" between the at least one output and the corresponding result data field, to upload to operational predictive model module **406**.

[0151] In certain embodiments, the one or more machine learning models may include one or more neural networks, such as a convolutional neural network, a deep learning neural network, or the like. The neural network may have one or more layers of nodes, and the model parameters adjusted during training may be respective weight values applied to one or more inputs to each node to produce a node output. In other words, the nodes in each layer may receive one or more inputs and apply a weight to each input to generate a node output. The node inputs to the first layer may correspond to the model input data fields, and the node outputs of the final layer may correspond to the at least one output of the model, intended to predict the at least one result data field.

[0152] One or more intermediate layers of nodes may be connected between the nodes of the first layer and the nodes of the final layer. As model trainer module **418** cycles through the training data sets **414**, model trainer module **418** applies a suitable backpropagation algorithm to adjust the weights in each node layer to minimize the error between the at least one output and the corresponding result data field. In this fashion, the machine learning model is trained to produce output that reliably predicts the corresponding result data field. Alternatively, the machine learning model may have any suitable structure.

[0153] In some embodiments, model trainer module **418** provides an advantage by automatically discovering and properly weighting complex, second- or third-order, and/or otherwise nonlinear interconnections between the model input data fields and the at least one output. Absent the machine learning model, such connections are unexpected and/or undiscoverable by human analysts.

[0154] In exemplary embodiments, operational predictive model module **406** may compare feedback, and may route a comparison result **422** generated by comparing prediction **420** to the feedback to a model updater module **424** of server computing device **150**. Model updater module **424** is configured to derive a correction signal **426** from comparison results **422** received for one or more predictions **420**, and to provide correction signal **426** to model trainer module **418** to enable updating or "re-training" of the at least one machine learning model to improve performance. The retrained at least one machine learning model **416** may be periodically re-uploaded to operational predictive model module **406**.

Exemplary Computer System

[0155] FIG. **5** illustrates an exemplary computer system **500** for implementing system **100** (shown in FIG. **1**). In the exemplary embodiment, computer system **500** is used for analyzing sensor data associated with a home **130** (shown in FIG. **1**) for home monitoring and/or generating recommendations for home **130**.

[0156] In the exemplary embodiment, user devices **140** are computers that include a web browser or a software application, which enables user devices **140** to communicate with server computing device **150** using the Internet, a local area network (LAN), or a wide area network (WAN). In some embodiments, user devices **140** are communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a LAN, a WAN, or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, a satellite connection, and a cable modem. User devices **140** can be any device capable of accessing a network, such as the Internet, including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, virtual headsets or glasses (e.g., AR (augmented reality), VR (virtual reality), MR (mixed reality), or XR (extended reality) headsets or glasses), chat bots, voice bots, ChatGPT bots or ChatGPT-based bots, or other web-based connectable equipment or mobile devices.

[0157] In the exemplary embodiment, IoT devices **110**, smart sensors **112**, and/or gas shutoff valves **114** include computers that may include a web browser or a software application, which enables IoT devices **110**, smart sensors **112**, and/or gas shutoff valves **114** to communicate with server computing device **150** using the Internet, a local area network (LAN), or a wide area network (WAN). In some embodiments, IoT devices **110**, smart sensors **112**, and/or gas shutoff valves **114** are communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a LAN, a WAN, or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, a satellite connection, and a cable modem.

[0158] IoT devices **110** and/or smart sensors **112** can be any device capable of accessing a network, such as the Internet, including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, virtual headsets or glasses (e.g., AR (augmented reality), VR (virtual reality), MR (mixed reality), or XR (extended reality) headsets or glasses), chat bots, voice bots, ChatGPT bots or ChatGPT-based bots, or other web-based connectable equipment or mobile devices. In the exemplary embodiment, IoT devices **110**, smart sensors **112**, and/or gas shutoff valves **114** as devices connected to the home network **205** (shown in FIG. **2**) that provide information about the home **130**.

[0159] In the exemplary embodiment, manufacturer servers **105** are computers that may include a web browser or a software application, which enables manufacturer servers **105** to communicate with associated source IoT devices **110** and the server computing device **150** using the Internet, a local area network (LAN), or a wide area network (WAN). In some embodiments, the manufacturer servers **105** are communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a LAN, a WAN, or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, a satellite connection, and a cable modem. The manufacturer servers **105** can be any device capable of accessing a network, such as the Internet, including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, virtual headsets or glasses (e.g., AR (augmented reality), VR (virtual reality), MR (mixed reality), or XR (extended reality) headsets or glasses), chat bots, voice bots, ChatGPT bots or ChatGPT-based bots, or other web-based connectable equipment or mobile devices.

[0160] In the exemplary embodiment, marketplace servers **240** are computers that may include a web browser or a software application, which enables marketplace servers **240** to communicate with associated the server computing device **150** using the Internet, a local area network (LAN), or a wide area network (WAN). In some embodiments, the marketplace servers **240** are communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a LAN, a WAN, or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, a satellite connection, and a cable modem. The marketplace servers **240** can be any device capable of accessing a network, such as the Internet, including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, virtual headsets or glasses (e.g., AR (augmented reality), VR (virtual reality), MR (mixed reality), or XR (extended reality) headsets or glasses), chat bots, voice bots, ChatGPT bots or ChatGPT-based bots, or other web-based connectable equipment or mobile devices.

[0161] In the exemplary embodiment, server computing device **150** is a computer that may include a web browser or a software application, which enables server computing device **150** to communicate with user devices **140** using the Internet, a local area network (LAN), or a wide area network (WAN). In some embodiments, the server computing device **150** is communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a LAN, a WAN, or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, a satellite connection, and a cable modem. The server computing device **150** can be any device capable of accessing a network, such as the Internet, including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, virtual headsets or glasses (e.g., AR (augmented reality), VR (virtual reality), MR (mixed reality), or XR (extended reality) headsets or glasses), chat bots, voice bots, ChatGPT bots or ChatGPT-based bots, or other web-based connectable equipment or mobile devices.

[0162] A database server **502** is communicatively coupled to a database **402** that stores data. In one embodiment, the database **402** is a database that includes home data, sensor data, property data, and/or recommendations. In some embodiments, the database **402** is stored remotely from the server computing device **150**. In some embodiments, the database **402** is decentralized. In the exemplary embodiment, a person can access the database **402** via user devices **140** by logging onto server computing device **150**.

Exemplary Client Device

[0163] FIG. **6** depicts an exemplary configuration of a client computer device shown in FIG. **5**, in accordance with one embodiment of the present disclosure. User computer device **602** may be operated by a user **601**. User computer device **602** may include, but is not limited to, user device **140**, IoT devices **110**, smart sensors **112**, IoT camera **115**, IoT thermostat **120**, IoT door lock **125**, (all shown in FIG. **1**), EM devices **304**, HVAC devices **314**, home network computer devices **316**, smart speaker devices **318**, home entertainment devices **320**, home security system **322**, smart home system **324**, home power management system **326**, and/or home car charging station **328** (all shown in FIG. **3**).

[0164] User computer device **602** may include a processor **605** for executing instructions. In some embodiments, executable instructions are stored in a memory area **610**. Processor **605** may include one or more processing units (e.g., in a multi-core configuration). Memory area **610** may be any device allowing information such as executable instructions and/or transaction data to be stored and retrieved. Memory area **610** may include one or more computer-readable media.

[0165] User computer device **602** may also include at least one media output component **615** for presenting information to user **601**. Media output component **615** may be any component capable of conveying information to user **601**. In some embodiments, media output component **615** may include an output adapter (not shown) such as a video adapter and/or an audio adapter. An output adapter may be operatively coupled to processor **605** and operatively couplable to an output device such as a display device (e.g., a cathode ray tube (CRT), liquid crystal display (LCD), light emitting diode (LED) display, or "electronic ink" display), an audio output device (e.g., a speaker or headphones), virtual headsets (e.g., AR (Augmented Reality), VR (Virtual Reality), or XR (eXtended Reality) headsets), and/or voice or chat bots.

[0166] In some embodiments, media output component **615** may be configured to present a graphical user interface (e.g., a web browser and/or a client application) to user **601**. A graphical user interface may include, for example, an online score viewing interface for viewing a home health score and determining more information about the home health score. In some embodiments, user computer device **602** may include an input device **620** for receiving input from user **601**. User **601** may use input device **620** to, without limitation, select a provider.

[0167] Input device **620** may include, for example, a keyboard, a pointing device, a mouse, a stylus, a touch sensitive panel (e.g., a touch pad or a touch screen), a gyroscope, an accelerometer, a position detector, a biometric input device, and/or an audio input device. A single component such as a touch screen may function as both an output device of media output component **615** and input device **620**.

[0168] User computer device **602** may also include a communication interface **625**, communicatively coupled to a remote device such as the server computing device **150** (shown in FIG. **1**) and/or the marketplace server **240** (shown in FIG. **2**). Communication interface **625** may include, for example, a wired or wireless network adapter and/or a wireless data transceiver for use with a mobile telecommunications network.

[0169] Stored in memory area **610** are, for example, computer-readable instructions for providing a user interface to user **601** via media output component **615** and, optionally, receiving and processing input from input device **620**. A user interface may include, among other possibilities, a web browser and/or a client application. Web browsers enable users, such as user **601**, to display and interact with media and other information typically embedded on a web page or a website from the server computing device **150** and/or the marketplace server **240**. A client application allows user **601** to interact with, for example, server computing device **150** and/or the marketplace server **240**. For example, instructions may be stored by a cloud service, and the output of the execution of the instructions sent to the media output component **615**.

[0170] Processor **605** executes computer-executable instructions for implementing aspects of the disclosure. In some embodiments, the processor **605** is transformed into a special purpose microprocessor by executing computer-executable instructions or by otherwise being programmed.

Exemplary Server Device

[0171] FIG. **7** depicts an exemplary configuration of a server computing device **700** shown in FIG. **1**, in accordance with one embodiment of the present disclosure. Server computer device **700** may include, but is not limited to, home controller **135**, server computing device **150** (shown in FIG. **1**), external data sources **215**, marketplace server **240** (both shown in FIG. **2**), home security system **322**, smart home system **324**, and/or home power management system **326**, (all shown in FIG. **3**). Server computer device **700** may also include a processor **705** for executing instructions. Instructions may be stored in a memory area **710**. Processor **705** may include one or more processing units (e.g., in a multi-core configuration).

[0172] Processor **705** may be operatively coupled to a communication interface **715** such that server computer device **700** is capable of communicating with a remote device such as another server computer device **700**. For example, communication interface **715** may receive requests from user device **140** via the Internet, as illustrated in FIG. **5**.

[0173] Processor **705** may also be operatively coupled to a storage device **734**. Storage device **734** may be any computer-operated hardware suitable for storing and/or retrieving data, such as, but not limited to, data associated with database **402** (shown in FIG. **4**). In some embodiments, storage device **734** may be integrated in server computer device **700**. For example, server computer device **700** may include one or more hard disk drives as storage device **734**.

[0174] In other embodiments, storage device **734** may be external to server computer device **700** and may be accessed by a plurality of server computer devices **700**. For example, storage device **634** may include a storage area network (SAN), a network attached storage (NAS) system, and/or multiple storage units such as hard disks and/or solid state disks in a redundant array of inexpensive disks (RAID) configuration.

[0175] In some embodiments, processor **705** may be operatively coupled to storage device **634** via a storage interface **720**. Storage interface **720** may be any component capable of providing processor **605** with access to storage device **734**. Storage interface **720** may include, for example, an Advanced Technology Attachment (ATA) adapter, a Serial ATA (SATA) adapter, a Small Computer System Interface (SCSI) adapter, a RAID controller, a SAN adapter, a network adapter, and/or any component providing processor **705** with access to storage device **734**.

[0176] Processor **705** may execute computer-executable instructions for implementing aspects of the disclosure. In some embodiments, the processor **705** may be transformed into a special purpose microprocessor by executing computer-executable instructions or by otherwise being programmed. For example, the processor **705** may be programmed with the instructions such as illustrated in FIG. **8**.

Exemplary Computer-Implemented Method for Gas Monitoring in a Home

[0177] FIGS. **8**A, **8**B, and **8**C depict a flow chart of an exemplary computer-implemented method **800** for gas monitoring in a home such as home **130** using system **100** (shown in FIG. **1**).

[0178] In the exemplary embodiment, method **800** may include receiving sensor data from at least one smart sensor (Block **802**). In some embodiments, the operations and/or functionality of Block **802** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

[0179] In some embodiments, method **800** may further include generating an artificial intelligence model based upon the historical sensor data relating to the plurality of homes (Block **804**). In some embodiments, the operations and/or functionality of Block **804** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

[0180] In the exemplary embodiment, method **800** may further include comparing the sensor data to one or more parameters generated using the artificial intelligence model to determine an abnormal level of gas is present in the home (Block **806**). In some embodiments, the operations and/or functionality of Block **806** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

[0181] In some embodiments, method **800** may further include comparing the sensor data to one or more baseline norms to determine an abnormal level of gas is present in the home (Block **808**). In some embodiments, the operations and/or functionality of Block **808** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

[0182] In some embodiments, method **800** may further include comparing the sensor data to the one or more the sensor data to the plurality of data values of the digital profiles to determine an abnormal level of gas is present in the home (Block **810**). In some embodiments, the operations and/or functionality of Block **810** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

[0183] In the exemplary embodiment, method **800** may further include, in response to determining that an abnormal level of gas is present in the home, controlling the at least one gas shutoff valve to close to prevent further gas leaks at the home (Block **812**). In some embodiments, the operations and/or functionality of Block **812** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

[0184] In some embodiment, method **800** may further include transmitting content data to a user device associated with the home that, when received by the user device, causes the user device to generate a user interface including an indicator that an abnormal level of gas is present in the home (Block **814**). In some embodiments, the operations and/or functionality of Block **814** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

[0185] In some embodiments, method **800** may further include identifying, using the one or more parameters, a location of a gas leak within the home (Block **816**). In some embodiments, the operations and/or functionality of Block **816** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

[0186] In some embodiments, method **800** may further include selecting at least one of a plurality of gas shutoff valves to close based upon the location of the gas leak (Block **818**). In some embodiments, the operations and/or functionality of Block **818** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

[0187] In some embodiments, method **800** may further include transmitting content data to the user device associated with the home that, when received by the user device, causes the user device to generate a user interface including the location of the gas leak (Block **820**). In some embodiments, the operations and/or functionality of Block **820** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

[0188] In some embodiments, method **800** may further include receiving further sensor data relating to the plurality of homes (Block **822**), updating the artificial intelligence model based upon the further sensor data (Block **824**), and updating the one or more parameters using the updated artificial intelligence model (Block **826**). In some embodiments, the operations and/or functionality of Block **822**, Block **824**, and/or Block **826** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

[0189] In some embodiments, method **800** may further include receiving, by a home controller, the one or more parameters from a server computing device (Block **828**). In some embodiments, the operations and/or functionality of Block **828** may be performed by server computing device **150** and/or home controller **135** (shown in FIG. **1**).

Exemplary Data Analysis Process for Home Monitoring

[0190] FIG. **9** is a diagram illustrating an exemplary data analysis process **900** for gas monitoring in a home that may be performed by system **100** (shown in FIG. **1**), for example, using server computing device **150** and/or home controller **135**.

[0191] Server computing device **150** and/or home controller **135** may collect information from smart sensors placed at various locations **902** within a home, which may include, for example: (1) any mechanicals or elements that have a natural gas or propane connection; (2) gas range; (3) gas oven; (4) gas dryer; (5) gas furnace or heater; (6) gas fireplace; an/or (7) gas water heater.

[0192] Server computing device **150** and/or home controller **135** may generate, based on sensor data detected at locations **902**, one or more data outputs **904**, which may include, for example: (1) smart controller syncs with smart sensors to pull in data, understand status, engage to perform action, and send alerts to the user; (2) if gas is detected at a single location, smart sensor detects presence of gas by an element or gas connection and smart shutoff closes the gas valve attached to the single element or gas connection; (3) if gas is detected at multiple locations, smart sensor detects presence of gas by two or more elements or gas connections or other smart sensors programmed to detect the digital signature of gas and smart shutoff closes valves closest to the presence of gas as well as the main gas valve to the structure; (4) alerts are sent to user(s) about presence of gas, location, and that shutoff was successful or unsuccessful; (5) user is provided with option to contact emergency services to shut off gas and perform inspection; (6) status of all sensors, alerts, actions to be taken, recommendations, history, reports can be provided on a home health monitor information platform app; (7) ongoing analysis of insurance claim data could proactively identify make/model of ranges or other appliances which may be defective and notify customers and manufacturers; and/or (8) overall monthly home health reports, overall monthly home health reports, annual reports, historical reports, reports for home buyers/sellers or insurance.

Machine Learning and Other Matters

[0193] The computer-implemented methods discussed herein may include additional, less, or alternate actions, including those discussed elsewhere herein. The methods may be implemented via one or more local or remote processors, transceivers, servers, and/or sensors (such as processors, transceivers, servers, and/or sensors mounted on vehicles or mobile devices, or associated with smart infrastructure or remote servers), and/or via computer-executable instructions stored on non-transitory computer-readable media or medium.

[0194] In some embodiments, server computing device **150** is configured to implement machine learning, such that server computing device **150** "learns" to analyze, organize, and/or process data without being explicitly programmed. Machine learning may be implemented through machine learning methods and algorithms ("ML methods and algorithms"). In an exemplary embodiment, a machine learning module ("ML module") is configured to implement ML methods and algorithms. In some embodiments, ML methods and algorithms are applied to data inputs and generate machine learning outputs ("ML outputs"). Data inputs may include but are not limited to images. ML outputs may include, but are not limited to identified objects, items classifications, and/or other data extracted from the images. In some embodiments, data inputs may include certain ML outputs.

[0195] In some embodiments, at least one of a plurality of ML methods and algorithms may be applied, which may include but are not limited to: linear or logistic regression, instance-based algorithms, regularization algorithms, decision trees, Bayesian networks, cluster analysis, association rule learning, artificial neural networks, deep learning, combined learning, reinforced learning, dimensionality reduction, and support vector machines. In various embodiments, the implemented ML methods and algorithms are directed toward at least one of a plurality of categorizations of machine learning, such as supervised learning, unsupervised learning, and reinforcement learning.

[0196] In one embodiment, the ML module employs supervised learning, which involves identifying patterns in existing data to make predictions about subsequently received data. Specifically, the ML module is "trained" using training data, which includes example inputs and associated example outputs. Based upon the training data, the ML module may generate a predictive function which maps outputs to inputs and may utilize the predictive function to generate ML outputs based upon data inputs. The example inputs and example outputs of the training data may include any of the data inputs or ML outputs described above. In the exemplary embodiment, a processing element may be trained by providing it with a large sample of home attributes with known characteristics or features. Such information may include, for example, sensor data and/or information associated with a plurality of IoT devices **110**.

[0197] In another embodiment, a ML module may employ unsupervised learning, which involves finding meaningful relationships in unorganized data. Unlike supervised learning, unsupervised learning does not involve user-initiated training based upon example inputs with associated outputs. Rather, in unsupervised learning, the ML module may organize unlabeled data according to a relationship determined by at least one ML method/algorithm employed by the ML module. Unorganized data may include any combination of data inputs and/or ML outputs as described above.

[0198] In yet another embodiment, a ML module may employ reinforcement learning, which involves optimizing outputs based upon feedback from a reward signal. Specifically, the ML module may receive a user-defined reward signal definition, receive a data input, utilize a decision-making model to generate a ML output based upon the data input, receive a reward signal based upon the reward signal definition and the ML output, and alter the decision-making model so as to receive a stronger reward signal for subsequently generated ML outputs. Other types of machine learning may also be employed, including deep or combined learning techniques.

[0199] In some embodiments, generative artificial intelligence (AI) models (also referred to as generative machine learning (ML) models) may be utilized with the present embodiments, and may the voice bots or chatbots discussed herein may be configured to utilize artificial intelligence and/or machine learning techniques. For instance, the voice or chatbot may be a ChatGPT chatbot. The voice or chatbot may employ supervised or unsupervised machine learning techniques, which may be followed by, and/or used in conjunction with, reinforced or reinforcement learning techniques. The voice or chatbot may employ the techniques utilized for ChatGPT. The voice bot, chatbot, ChatGPT-based bot, ChatGPT bot, and/or other bots may generate audible or verbal output, text or textual output, visual or graphical output, output for use with speakers and/or display screens, and/or other types of output for user and/or other computer or bot consumption.

[0200] Based upon these analyses, the processing element may learn how to identify characteristics and patterns that may then be applied to analyzing and classifying objects. The processing element may also learn how to identify attributes of different objects in different lighting. This information may be used to determine which classification models to use and which classifications to provide.

Exemplary Embodiments

[0201] In an exemplary embodiment, a computer system for monitoring of a home may be provided. The computer system may include at least one memory device, at least one processor in communication with the at least one memory device, at least one smart sensor for the home, and at least one gas shutoff valve of the home. The computer system may be programmed to: (a) receive sensor data from at least one smart sensor; (b) compare the sensor data to one or more parameters generated using an artificial intelligence model to determine that an abnormal level of gas is present in the home, wherein the artificial intelligence model is trained based upon historical sensor data relating to a plurality of homes; and/or (c) in response to determining that an abnormal level of gas is present in the home, control at least one gas shutoff valve to close to prevent further gas leaks at the home. The computer system may include additional, less, or alternate functionality, including that discussed elsewhere herein.

[0202] In another exemplary embodiment, a computer-implemented method for monitoring of a home may be provided. The computer-implemented method may be performed by a computing device including at least one processor and at least one memory device. The method may include, via the at least one processor: (a) receiving sensor data from at least one smart sensor; (b) comparing the sensor data to one or more parameters generated using an artificial intelligence model to determine an abnormal level of gas is present in the home, wherein the artificial intelligence model is trained based upon historical sensor data relating to a plurality of homes; and/or (c) in response to determining that an abnormal level of gas is present in the home, controlling at least one gas shutoff valve to close to prevent further gas leaks at the home. The method may have additional, less, or alternate actions, including that discussed elsewhere herein.

[0203] In yet another exemplary embodiment, at least one non-transitory computer-readable medium having computer-executable instructions embodied thereon may be provided. When executed by at least one processor, the computer-executable instructions cause the at least one processor a to: (a) receive sensor data from at least one smart sensor; (b) compare the sensor data to one or more parameters generated using an artificial intelligence model to determine an abnormal level of gas is present in the home, wherein the artificial intelligence model is trained based upon historical sensor data relating to a plurality of homes; and/or (c) in response to determining that an abnormal level of gas is present in the home, control at least one gas shutoff valve to close to prevent further gas leaks at the home. The computer-readable medium may have instructions that direct additional, less, or alternate functionality, including that discussed elsewhere herein.

Additional Considerations

[0204] As will be appreciated based upon the foregoing specification, the above-described embodiments of the disclosure may be implemented using computer programming or engineering techniques including computer software, firmware, hardware or any combination or subset thereof. Any such resulting program, having computer-readable code means, may be embodied or provided within one or more computer-readable media, thereby making a computer program product, i.e., an article of manufacture, according to the discussed embodiments of the disclosure. The computer-readable media may be, for example, but is not limited to, a fixed (hard) drive, diskette, optical disk, magnetic tape, semiconductor memory such as read-only memory (ROM), and/or any transmitting/receiving medium such as the Internet or other communication network or link. The article of manufacture containing the computer code may be made and/or used by executing the code directly from one medium, by copying the code from one medium to another medium, or by transmitting the code over a network.

[0205] These computer programs (also known as programs, software, software applications, "apps," or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms "machine-readable medium" "computer-readable medium" refers to any computer program product, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The "machine-readable medium" and "computer-readable medium," however, do not include transitory signals. The term "machine-readable signal" refers to any signal used to provide machine instructions and/or data to a programmable processor.

[0206] As used herein, the term "database" can refer to either a body of data, a relational database management system (RDBMS), or to both. As used herein, a database can include any collection of data including hierarchical databases, relational databases, flat file databases, object-relational databases, object-oriented databases, and any other structured collection of records or data that is stored in a computer system. The above examples are example only, and thus are not intended to limit in any way the definition and/or meaning of the term database. Examples of RDBMS' include, but are not limited to including, Oracle® Database, MySQL, IBM® DB2, Microsoft® SQL Server, Sybase®, and PostgreSQL. However, any database can be used that enables the systems and methods described herein. (Oracle is a registered trademark of Oracle Corporation, Redwood Shores, California; IBM is a registered trademark of International Business Machines Corporation, Armonk, New York; Microsoft is a registered trademark of Microsoft Corporation, Redmond, Washington; and Sybase is a registered trademark of Sybase, Dublin, California.)

[0207] As used herein, a processor may include any programmable system including systems using micro-controllers, reduced instruction set circuits (RISC), application specific integrated circuits (ASICs), logic circuits, and any other circuit or processor capable of executing the functions described herein. The above examples are example only, and are thus not intended to limit in any way the definition and/or meaning of the term "processor."

[0208] As used herein, the terms "software" and "firmware" are interchangeable, and include any computer program stored in memory for execution by a processor, including RAM memory, ROM memory, EPROM memory, EEPROM memory, and non-volatile RAM (NVRAM) memory. The above memory types are example only, and are thus not limiting as to the types of memory usable for storage of a computer program.

[0209] In another example, a computer program is provided, and the program is embodied on a computer-readable medium. In an example, the system is executed on a single computer system, without requiring a connection to a server computer. In a further example, the system is being run in a Windows® environment (Windows is a registered trademark of Microsoft Corporation, Redmond, Washington). In yet another example, the system is run on a mainframe environment and a UNIX® server environment (UNIX is a registered trademark of X/Open Company Limited located in Reading, Berkshire, United Kingdom). In a further example, the system is run on an iOS® environment (iOS is a registered trademark of Cisco Systems, Inc. located in San Jose, CA). In yet a further example, the system is run on a Mac OS® environment (Mac OS is a registered trademark of Apple Inc. located in Cupertino, CA). In still yet a further example, the system is run on Android® OS (Android is a registered trademark of Google, Inc. of Mountain View, CA). In another example, the system is run on Linux® OS (Linux is a registered trademark of Linus Torvalds of Boston, MA). The application is flexible and designed to run in various different environments without compromising any major functionality.

[0210] In some embodiments, the system includes multiple components distributed among a plurality of computing devices. One or more components may be in the form of computer-executable instructions embodied in a computer-readable medium. The systems and processes are not limited to the specific embodiments described herein. In addition, components of each system and each process can be practiced independent and separate from other components and processes described herein. Each component and process can also be used in combination with other assembly packages and processes.

[0211] As used herein, an element or step recited in the singular and proceeded with the word "a" or "an" should be understood as not excluding plural elements or steps, unless such exclusion is explicitly recited. Furthermore, references to "example" or "one example" of the present disclosure are not intended to be interpreted as excluding the existence of additional examples that also incorporate the recited features. Further, to the extent that terms "includes," "including," "has," "contains," and variants thereof are used herein, such terms are intended to be inclusive in a manner similar to the term "comprises" as an open transition word without precluding any additional or other elements.

[0212] Furthermore, as used herein, the term "real-time" refers to at least one of the time of occurrence of the associated events, the time of measurement and collection of predetermined data, the time to process the data, and the time of a system response to the events and the environment. In the examples described herein, these activities and events occur substantially instantaneously.

[0213] The patent claims at the end of this document are not intended to be construed under 35 U.S.C. § 112(f) unless traditional means-plus-function language is expressly recited, such as "means for" or "step for" language being expressly recited in the claim(s).

[0214] This written description uses examples to disclose the disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

What is claimed is:

1. A computer system for monitoring levels of gases in a home, the computer system comprising: at least one memory device, at least one processor in communication with the at least one memory device, at least one smart sensor for the home, and at least one gas shutoff valve of the home, the at least one processor configured to:

receive sensor data from the at least one smart sensor;

compare the sensor data to one or more parameters generated using an artificial intelligence model to determine an abnormal level of gas is present in the home, wherein the artificial intelligence model is trained based upon historical sensor data relating to a plurality of homes; and

in response to determining that an abnormal level of gas is present in the home, control the at least one gas shutoff valve to close to prevent further gas leaks at the home.

2. The computer system of claim **1**, wherein the at least one processor is further configured to transmit content data to a user device associated with the home that, when received by the user device, causes the user device to generate a user interface including an indicator that an abnormal level of gas is present in the home.

3. The computer system of claim **1**, wherein the at least one processor is further configured to identify, using the one or more parameters, a location of a gas leak within the home.

4. The computer system of claim **3**, wherein the at least one processor is in communication with a plurality of gas shutoff valves of the home, and wherein the at least one processor is further configured to select at least one of the plurality of gas shutoff valves to close based upon the location of the gas leak.

5. The computer system of claim **3**, wherein the at least one processor is further configured to transmit content data to a user device associated with the home that, when received by the user device, causes the user device to generate a user interface including the location of the gas leak.

6. The computer system of claim **1**, wherein the at least one processor is further configured to generate the artificial intelligence model based upon the historical sensor data relating to the plurality of homes.

7. The computer system of claim **6**, wherein the at least one processor is further configured to:

receive further sensor data relating to the plurality of homes;

update the artificial intelligence model based upon the further sensor data; and

update the one or more parameters using the updated artificial intelligence model.

8. The computer system of claim **1**, further comprising a home controller disposed in the home and including the at least one processor, wherein the at least one processor is further configured to receive the one or more parameters from a server computing device.

9. The computer system of claim **1**, wherein the parameters include one or more baseline norms, and wherein the at least one processor is configured to compare the sensor data to the one or more baseline norms to determine an abnormal level of gas is present in the home.

10. The computer system of claim **1**, wherein the parameters include one or more digital profiles including a plurality of data values corresponding to different types of sensor data, and wherein the at least one processor is configured to compare the sensor data to the plurality of data values of the digital profiles to determine an abnormal level of gas is present in the home.

11. A computer-implemented method for gas monitoring in a home, the computer-implemented method performed by a computer system including at least one memory device and at least one processor in communication with the at least one memory device, at least one smart sensor for a home, and at least one gas shutoff valve of the home, the computer-implemented method comprising:

receiving sensor data from the at least one smart sensor;

comparing the sensor data to one or more parameters generated using an artificial intelligence model to determine an abnormal level of gas is present in the home, wherein the artificial intelligence model is trained based upon historical sensor data relating to a plurality of homes; and

in response to determining that an abnormal level of gas is present in the home, controlling the at least one gas shutoff valve to close to prevent further gas leaks at the home.

12. The computer-implemented method of claim **11**, further comprising transmitting content data to a user device associated with the home that, when received by the user device, causes the user device to generate a user interface including an indicator that an abnormal level of gas is present in the home.

13. The computer-implemented method of claim **11**, further comprising identifying, using the one or more parameters, a location of a gas leak within the home.

14. The computer-implemented method of claim **13**, wherein the at least one processor is in communication with a plurality of gas shutoff valves of the home, and the computer-implemented method further comprises selecting at least one of the plurality of gas shutoff valves to close based upon the location of the gas leak.

15. The computer-implemented method of claim **13**, further comprising transmitting content data to a user device associated with the home that, when received by the user device, causes the user device to generate a user interface including the location of the gas leak.

16. The computer-implemented method of claim **11**, further comprising generating the artificial intelligence model based upon the historical sensor data relating to the plurality of homes.

17. The computer-implemented method of claim **16**, further comprising:

receiving further sensor data relating to the plurality of homes;

updating the artificial intelligence model based upon the further sensor data; and

updating the one or more parameters using the updated artificial intelligence model.

18. The computer-implemented method of claim **11**, wherein the computer system further includes a home controller disposed in the home and including the at least one processor, and wherein the computer-implemented method further comprises receiving by the home controller the one or more parameters from a server computing device.

19. The computer-implemented method of claim **11**, wherein the parameters include one or more baseline norms, and wherein the computer-implemented method further comprises comparing the sensor data to the one or more baseline norms to determine an abnormal level of gas is present in the home.

20. The computer-implemented method of claim **11**, wherein the parameters include one or more digital profiles including a plurality of data values corresponding to different types of sensor data, and wherein the computer-implemented method further comprises comparing the sensor data to the plurality of data values of the digital profiles to determine an abnormal level of gas is present in the home.

21. At least one non-transitory computer-readable media having computer-executable instructions embodied thereon, wherein when executed by a computer system including at least one memory device and at least one processor in communication with the at least one memory device, at least one smart sensor for a home, and at least one gas shutoff valve of the home, the computer-executable instructions cause the at least one processor to:

receive sensor data from the at least one smart sensor;

compare the sensor data to one or more parameters generated using an artificial intelligence model to determine an abnormal level of gas is present in the home, wherein the artificial intelligence model is trained based upon historical sensor data relating to a plurality of homes; and

in response to determining that an abnormal level of gas is present in the home, control the at least one gas shutoff valve to close to prevent further gas leaks at the home.

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